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Variable frequency-based multi-frame coherent track-before-detect method for weak tones in passive sonar

Liu Zhang,^{1,2} Shengchun Piao,^{2,3,4,a)}  Junyuan Guo,^{2,3,4} and Xiaohan Wang^{2,3,4} 

¹Hanjiang National Laboratory, Wuhan, Hubei 430060, China

²College of Underwater Acoustic Engineering, Harbin Engineering University, Harbin 150001, China

³National Key Laboratory of Underwater Acoustic Technology, Harbin Engineering University, Harbin 150001, China

⁴Key Laboratory of Marine Information Acquisition and Security (Harbin Engineering University), Ministry of Industry and Information Technology, Harbin 150001, China

ABSTRACT:

Passive detection for weak tones remains a challenging topic. Tonal frequency trajectory can be extracted by combining the pre-processing based on the multi-frame coherent integration with track-before-detect (TBD) method. However, complex target maneuvers can lead to intricate variations in tonal frequency, limiting the coherent processing gain. To address this issue, a variable frequency-based multi-frame coherent track-before-detect method is proposed. The evolution of tonal frequency and phase across time frames is modeled using polynomial functions. We propose a state-space dynamical system model for the time-evolving tonal signal, where the state variables are defined as the tonal amplitude and the coefficients of the polynomial used to represent the tonal frequency. The optimal model order is then analyzed based on minimizing the coherent gain loss. Furthermore, an improved particle filtering algorithm is employed to implement the established TBD model. We design a data-adaptive sequential importance sampling method. By optimizing particle sampling based on high transition probabilities, a majority of particles can be distributed in high-likelihood regions. This enables high adaptability when the tonal frequency undergoes complex variations. Both simulation and processing results from SwellEx-96 experiment demonstrate that the proposed method can improve detection performance and reduce frequency estimation error.

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I. INTRODUCTION

Narrowband line spectral components^{1,2} (also known as tones) are crucial features for underwater passive sonar detection,^{3,4} tracking,⁵ and recognition.⁶ In underwater unmanned combat scenarios, large aperture sonar arrays are not feasible due to limited platform size. It is more common to deploy a single sensor.⁷ With advancements in vibration and noise reduction technologies and the increasing noise levels in the ocean environment, the tonal signal-to-noise ratio (SNR) is gradually decreasing.⁸ This poses greater demands on single-channel signal processing methods. Moreover, the Doppler effect due to target motion leads to time variations in the tonal frequency. This significantly limits the effective integration duration, resulting in a substantial reduction in processing gain.⁹ Therefore, there is a strong motivation to develop more robust methods for detecting weak tones with varying frequencies.

Low frequency analysis and recording (LOFAR) is one of the commonly used techniques for analyzing time-varying tones.¹⁰ Its main principle involves transforming a tone from time-domain into time-frequency domain through segmented discrete Fourier transform (DFT) operations. The

spectrum for each time segment (also called frame) in the LOFAR depicts the time-varying characteristics of the tonal frequency visually. Scholars have proposed many detection and tracking algorithms based on LOFAR technique. They can be roughly categorized into two types. The first type consists of detect-before-track (DBT) methods. In these methods, a threshold^{11,12} is applied to the spectrum at each frame in the LOFAR, and the frequency bins that exceed the threshold are treated as tone candidates. The detected tone candidates are then linked through association algorithms¹³ to establish a continuous frequency trajectory. The second type consists of track-before-detect (TBD) methods.¹⁴ The core principle involves initially reducing the detection threshold used to detect frequency bins eligible for tracking, acknowledging a temporary elevation in the false alarm probability, knowing that the subsequent tracking (association of such bins to existing tracks or birth of new tracks) will eliminate most of the false alarms, when tracks are pruned. Tracks are treated as hypothetical only until enough detected bins have been added to them to raise their confidence. Consequently, this strategy of lowering the detection threshold increases the system sensitivity^{15,16} compared with the conventional DBT methods.

Motivated by the TBD concept, some researchers have proposed using the analytical expression of the spectrum to

^{a)}Email: piaoshengchun@hrbeu.edu.cn

establish the dynamic model and measurement model. The extended Kalman filter (EKF) algorithm was then employed to estimate the frequency state.^{17,18} However, this algorithm is only effective under high SNR conditions. The hidden Markov model (HMM) allows for modeling the unobservable frequency state.^{19,20} By utilizing a batch processing strategy to estimate the maximum likelihood path of the frequency state from tonal amplitude observations,^{21–23} the tonal frequency trajectory can be extracted. Furthermore, the particle filtering (PF) method, known for its powerful capabilities in handling nonlinear and non-Gaussian problems,²⁴ has also been employed for detecting weak tones. Zhang *et al.*²⁵ established a TBD model and likelihood ratio function based on the amplitude statistical characteristics of single-frame spectra track-before-detect (SF-TBD) method. This approach allows for flexible estimation of the existence states for multiple tones. However, the aforementioned methods rely only on amplitude information from single-frame measurements, discarding phase information. This omission could potentially hinder their ability to detect tones under lower SNR.

The methods that utilize phase relationships between tones to enhance detection performance can be referred to as coherent processing. Davey *et al.*²⁶ developed an intra-frame coherent TBD model based on the phase relationships between individual pixels within a single-frame radar image. Compared to TBD models that do not incorporate phase information, this approach significantly enhances the capability of detecting weak targets.^{27,28} Barrett and Holdsworth²⁹ designed a likelihood ratio function based on the tonal phase information between adjacent frames for HMM model, achieving higher frequency estimation accuracy. Suvorova *et al.*³⁰ and Carevic,³¹ respectively, utilized the phase-wrapped Ornstein–Uhlenbeck process and continuous-time posterior distribution to describe the tonal states' evolution between adjacent frames, allowing for simultaneous estimation of both the frequency and phase. Compared to methods utilizing intra-frame phase, the approaches that leverage inter-frame phase can effectively enhance detection capabilities. However, only measurements from two adjacent frames are considered in these methods, and the phase estimation process is unavoidable, which may reduce the robustness of the algorithm.

To address this, we have previously proposed a multi-frame coherent track-before-detect (MFC-TBD) method.³² This method allows coherent accumulation of tonal information over multiple frames within a single batch, greatly improving the SNR of the tone. The parallel execution of the PF across multiple frequency bins enables the simultaneous detection of multiple tone signals. However, the assumption of constant frequency within a single batch leads to a significant performance degradation in the presence of frequency variations. Furthermore, complex maneuvering of the target will result in intricate variations in tonal frequency. When the actual frequency trajectory deviates from the assumed model, a majority of predicted particles will be distributed in regions away from the true frequency state,

leading to degraded filtering performance. Auxiliary particle filtering (APF)³³ and PF methods based on intelligent optimization concepts³⁴ can mitigate this impact to some extent. However, the APF algorithm requires selecting effective particles based on current measurements, limiting its applicability in low SNR scenarios. Intelligent PF methods, while effective, can disrupt the distribution characteristics of the prediction function, necessitating additional weight updates and compensation operations, which increase the algorithm complexity.

In this study, we propose a variable frequency-based multi-frame coherent track-before-detect (VFMFC-TBD) method to enhance the detection performance of weak tones with varying frequency. Abandoning the assumption of constant tone frequency, this method models tonal phase and frequency as polynomials within a batch over a number of Fourier epochs, enabling the development of a novel state-space dynamical system model based on batch processing. To address the performance degradation of conventional PF algorithm in the presence of complex variations in tonal frequency, a data-adaptive sequential importance sampling (DA-SIS) is designed. The proposed strategy can concentrate the majority of predicted particles in the high-likelihood region, leading to increased particle utilization efficiency. The detection and tracking performance of the single-frame track-before-detect (SF-TBD),²⁵ multi-frame non-coherent track-before-detect (MFNC-TBD),³⁵ multi-frame coherent track-before-detect (MFC-TBD),³² and the proposed VFMFC-TBD methods are compared through simulation and the real data from SwellEx-96 experiment. This paper is organized as follows: The signal model and problem of the current research are formulated in Sec. II. Section III introduces the principle of the proposed VFMFC-TBD method. Simulation and experimental results are presented in Secs. IV and V, respectively. The conclusions are given in Sec. VI.

II. PROBLEM FORMULATION

A. Signal model in time domain

While there is extensive literature on modeling received signals from targets undergoing uniform linear motion, too little research focuses on targets executing arbitrary motion (curvilinear or non-uniform). An approximate expression for the received signal is presented by employing a discretization approach in this section, which consists of three steps: (1) The target trajectory is first divided into a finite number of short trajectories based on a specific time interval. Within each short trajectory, the instantaneous frequency of the target is approximated as constant. (2) Subsequently, the phase of the received signal at each discrete time instant is obtained by integrating the instantaneous frequency over each time interval. (3) Finally, these discrete phases are concatenated to yield the expression for the received signal throughout the trajectory.

As depicted in Fig. 1, assume a stationary receiver situated at the origin of a Cartesian coordinate system. A

moving target equipped with a source emits a pure tone signal. The target trajectory is represented by a blue dashed line, while its velocity magnitude and direction are indicated by vector $\vec{v}$. Let (x, y) represent the coordinates of any point on the target trajectory. The angle between the line connecting this point to the origin and the positive y -axis is denoted as β . In addition, $\cos \beta = y/\sqrt{x^2 + y^2}$, $\sin \beta = x/\sqrt{x^2 + y^2}$. The target velocity components in the x and y directions are denoted as v_x and v_y , respectively. v_r represents the target radial velocity, $v_r = v_x \sin \beta + v_y \cos \beta$, as described in the velocity triangle inset.

The target radial velocity at time m can be written as $v_{r,m} = v_{x,m} \sin \beta_m + v_{y,m} \cos \beta_m$, where m represents the index of the target short trajectory. Each target short trajectory can be approximated as a straight line, with a constant radial velocity $v_{r,m}$. Then, the instantaneous frequency of the received signal at time m , due to Doppler effect, is^{2,36}

$$f_m = f_0 - \frac{v_{x,m} \sin \beta_m + v_{y,m} \cos \beta_m}{c}, \quad (1)$$

where f_0 is the source frequency, c represents the speed of sound in water.

Finally, the theoretical expression for the received signal, derived from the relationship between instantaneous frequency and phase, can be written as

$$s_r(n) = A(n) \exp \left[j \left(\sum_{m=1}^n 2\pi f_m T_s + \varphi_0 \right) \right], \quad n = 0, 1, \dots, N-1, \quad (2)$$

where n represents the discrete sample index in the time domain, $A(n)$ represents the amplitude of the received signal, which may vary with time, T_s represents the sampling interval, and φ_0 is the initial phase.

B. Limitations and motivations

The core of the MFC-TBD method described in Ref. 32 is the coherent accumulation of the spectra located in the same discrete frequency bins between multiple frames

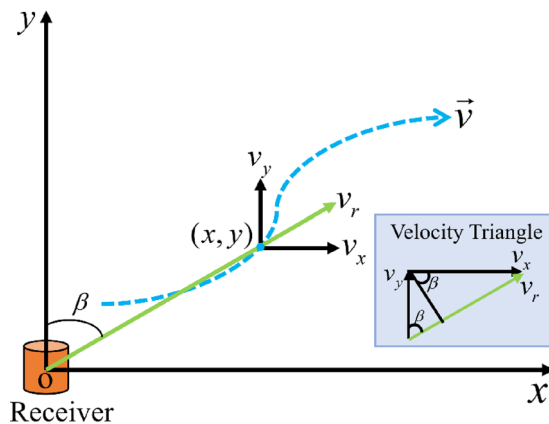


FIG. 1. Target-receiver geometry.

within a single batch. For the MFC-TBD method to be successful, the frame length T_0 and batch length W must be chosen such that

$$\left| \int_{t_k}^{t_k + WT_0} \dot{f}(t) dt \right| < \Delta f, \quad (3)$$

where $\dot{f}(t)$ denotes the rate of change of the tonal instantaneous frequency, $\Delta f = 1/T_0$ is the width of the discrete frequency bin, and t_k represents the starting time of the k th batch.

However, the condition in Eq. (3) is frequently violated in numerous underwater target motion scenarios. For targets in uniform linear motion, larger batch lengths, source velocities, and source frequencies correlate with greater rates of change, potentially causing the frequency variation within a single batch to exceed Δf . In this case, the signal energy in different frames will be concentrated in different frequency bins. The performance of the MFC-TBD method will be severely degraded or even fail. Therefore, developing novel TBD method becomes imperative and urgent to overcome this limitation.

III. VAEIABLE FREQUENCY-BASED MULTI-FRAME COHERENT TRACK-BEFORE DETECT METHOD

A. TBD model

By virtue of the Taylor series expansion theorem, the theoretical signal model in Eq. (2) can be approximated as a polynomial phase signal of order M . Thus, the observation sequence of the passive sonar can be represented as

$$s(n) = A(n)e^{j\phi_n(n)} + g(n), \quad n = 0, 1, \dots, N-1. \quad (4)$$

The instantaneous phase $\phi_t(n)$ and instantaneous frequency $f_t(n)$ of the signal are written as

$$\begin{aligned} \phi_t(n) &= 2\pi \sum_{m=0}^M \mu_m (t_n)^m, \\ f_t(n) &= \frac{1}{2\pi} \frac{d\phi}{dt} = \sum_{m=1}^M \mu_m m (t_n)^{m-1}, \\ t_n &= nT_s, \end{aligned} \quad (5)$$

where μ_m , $m = 0, 1, \dots, M$ are real coefficients that depend on the target's motion parameters. A closed-form solution for μ_m is generally inaccessible unless the target exhibits a highly predictable motion profile, such as uniform linear motion. $g(n)$ stands for Gaussian white noise with a mean of zero and variance of σ^2 .

Let the signal be segmented into K non-overlapping time frames, each containing L samples. The DFT result of the k th frame can be approximated as³⁷

$$\begin{aligned} Y_k(l) &= \frac{A_k \text{sinc}[\pi t_L (f_t(kL) - f_i)]}{\text{sinc}[\pi t_1 (f_t(kL) - f_i)]} \\ &\times e^{j\pi t_{L-1} [f_t(kL) - f_i]} e^{j\phi_t(kL)} + G_k(l), \end{aligned} \quad (6)$$

where $f_l = l/LT_s$, $l = 0, 1, \dots, L-1$, l represents the discrete frequency index. Equation (6) describes how the l th complex DFT bin output varies as the underlying signal frequency varies in neighborhood of f_l , the l th DFT bin frequency. $G_k(l)$ is the DFT result of k th noise segment. This approximation is valid when the maximum frequency variation within each segment is less than or equal to the width of the discrete frequency bins,

$$\left| \sum_{\lambda=2}^Y \lambda \mu_{\lambda} (MT_s)^{\lambda-1} \right| \leq \frac{1}{MT_s}. \quad (7)$$

1. Dynamic model

The unknown parameters of the tonal signal to be detected in Eq. (6) are A_k and μ_m , $m = 0, 1, \dots, M$, where A_k

represents the tonal amplitude and μ_m , $m = 1, 2, \dots, M$ are coefficients of an $M-1$ th-order polynomial used to represent the tonal frequency. These polynomial coefficients capture the temporal behavior of the frequency. $2\pi\mu_0$ represents the signal initial phase which is assumed to be uniformly distributed over $[0, 2\pi]$. Later, we will apply marginalization by integrating out the initial phase $2\pi\mu_0$. As such, the tonal state vector at time k is defined as

$$X_k = [A_k, \mu_{1,k}, \mu_{2,k}, \dots, \mu_{M,k}]^T. \quad (8)$$

The dynamic model is given by

$$X_k = FX_{k-1} + V_{k-1}, \quad (9)$$

where F represents the state transition matrix, expressed as

$$F = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 2T_0 & \dots & (M-1)(T_0)^{M-2} & M(T_0)^{M-1} \\ 0 & 0 & 1 & \dots & (M-1)(M-2)(T_0)^{M-3} & M(M-1)(T_0)^{M-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & \prod_{m=0}^{M-2} (M-m)T_0 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{bmatrix}. \quad (10)$$

The state noise vector V_k is given by

$$V_k = [q_A, q_{\mu_1}, q_{\mu_2}, \dots, q_{\mu_M}]^T. \quad (11)$$

The elements are random dynamic noise inputs for the state variable A_k and μ_m , $m = 1, 2, \dots, M$. The covariance matrix for V_k is written as

$$Q_{k-1} = \text{diag}(\sigma_A^2, \sigma_{\mu_1}^2, \sigma_{\mu_2}^2, \dots, \sigma_{\mu_M}^2), \quad (12)$$

where $\text{diag}(\cdot)$ represents a diagonal matrix, each diagonal element corresponds to the variance of the respective state noise variable in Eq. (11). Generally, the state noise variable follows a zero-mean Gaussian distribution.

To account for the presence or absence of tone, a binary variable e_k is introduced, where $e_k = 1$ indicates the presence and $e_k = 0$ indicates the absence. The evolution of e_k follows a Markov model with a transition probability matrix,

$$\Pi = \begin{bmatrix} 1 - P_b & P_b \\ P_d & 1 - P_d \end{bmatrix}, \quad (13)$$

where P_b stands for the tonal-birth probability, and P_d stands for the tonal-death probability.

2. Measurement model

The target SNR can be effectively improved by utilizing the information of multiple frames. Defining the measurement in κ th batch as a stacked vector of the DFT spectrum from time κ to $\kappa + W - 1$, denoted as $\mathbf{Z}_{\kappa}^{VFMFC}(l) = [Y_{\kappa}(l), \dots, Y_{\kappa+W-1}(l)]^T$, the measurement model can be expressed as

$$\mathbf{Z}_{\kappa}^{VFMFC}(l) = \begin{cases} \exp(j2\pi\mu_0)\mathbf{h}(X_{\kappa}) + \mathbf{G}_{\kappa}(l), & e_k = 1, \\ \mathbf{G}_{\kappa}(l), & e_k = 0, \end{cases} \quad (14)$$

where $\mathbf{h}(X_{\kappa}) = [h(X_{\kappa}), \dots, h(X_{\kappa+W-1})]^T$, $\mathbf{G}_{\kappa}(l) = [G_{\kappa}(l), \dots, G_{\kappa+W-1}(l)]^T$ and $\exp(j2\pi\mu_0)$ refers to the common initial phase of the signal spectra in each batch. According to Eq. (6), $h(X_{\kappa})$ is given by

$$h(X_{\kappa}) = \frac{A_{\kappa} \text{sinc}[\pi t_L(f_{\text{tr}}(kL) - f_l)]}{\text{sinc}[\pi t_1(f_{\text{tr}}(kL) - f_l)]} e^{j\pi t_{L-1}[f_{\text{tr}}(kL) - f_l]} e^{j\phi_{\text{tr}}(kL)}. \quad (15)$$

3. Likelihood ratio function

Based on the measurement model, the joint likelihood ratio function across multiple frames can be derived. To

facilitate the derivation process, the following variables will be simplified: $\mathbf{Z}_k^{VFMFC}(l) \rightarrow \mathbf{Z}$, $\mathbf{h}(\mathbf{X}_k) \rightarrow \mathbf{h}$, $e_k \rightarrow e$, $\mathbf{X}_k \rightarrow \mathbf{X}$, $\exp(j2\pi\mu_0) = \zeta$.

When the tone is present (i.e., $e = 1$), the spectrum of k th frame is characterized by a complex Gaussian distribution, i.e., $Y_k(l) \sim CN[h(\mathbf{X}_k), 2\sigma_G^2]$, where $\sigma_G^2 = \sigma^2/2M$. Estimation of σ^2 , the variance of the additive white Gaussian noise defined in Eq. (4), is performed using a maximum likelihood estimator developed in our prior work.³² The probability density of the measurement vector $\mathbf{Z}$ is expressed as

$$p(\mathbf{Z}|\mathbf{X}, e = 1) = \frac{1}{\sqrt{2\pi\sigma_G^{2W}}} \exp \left\{ -\frac{1}{2\sigma_G^2} [\mathbf{Z} - \zeta\mathbf{h}]^H [\mathbf{Z} - \zeta\mathbf{h}] \right\}. \quad (16)$$

When the tone is absent (i.e., $e = 0$), the spectrum of k th frame follows a complex Gaussian distribution with mean of 0, i.e., $Y_k(l) \sim CN(0, 2\sigma_G^2)$. The probability density of $\mathbf{Z}$ is written as

$$p(\mathbf{Z}|\mathbf{X}, e = 0) = \frac{1}{\sqrt{2\pi\sigma_G^{2W}}} \exp \left\{ -\frac{1}{2\sigma_G^2} \mathbf{Z}^H \mathbf{Z} \right\}. \quad (17)$$

According to Eqs. (16) and (17), the likelihood ratio function is constructed as

$$\begin{aligned} l(\mathbf{Z}|\mathbf{X}, e) &= \frac{p(\mathbf{Z}|\mathbf{X}, e = 1)}{p(\mathbf{Z}|\mathbf{X}, e = 0)} \\ &= \exp \left\{ -\frac{1}{2\sigma_G^2} [\mathbf{Z} - \zeta\mathbf{h}]^H [\mathbf{Z} - \zeta\mathbf{h}] + \frac{1}{2\sigma_G^2} \mathbf{Z}^H \mathbf{Z} \right\} \\ &= \exp \left\{ -\frac{1}{2\sigma_G^2} \mathbf{h}^H \mathbf{h} \right\} \\ &\quad \times \exp \left\{ \frac{1}{2\sigma_G^2} [\zeta\mathbf{Z}^H \mathbf{h} + \zeta^* \mathbf{h}^H \mathbf{Z}] \right\}. \end{aligned} \quad (18)$$

Through algebraic manipulation and marginalization over $2\pi\mu_0$, we obtain²⁶

$$\begin{aligned} l(\mathbf{Z}_k^{VFMFC}(l)|\mathbf{X}_k, e_k) &= \exp \left\{ -\frac{1}{2\sigma_G^2} \mathbf{h}^H(\mathbf{X}_k) \mathbf{h}(\mathbf{X}_k) \right\} I_0 \\ &\quad \times \left(\frac{|\mathbf{h}^H(\mathbf{X}_k) \mathbf{Z}_k^{VFMFC}(l)|}{\sigma_G^2} \right). \end{aligned} \quad (19)$$

In Eq. (19), the Bessel function plays a dominant role in determining the overall magnitude of the likelihood ratio function. The only significantly varying term within the Bessel function is $|\mathbf{h}^H(\mathbf{X}_k) \mathbf{Z}_k^{VFMFC}(l)|$, and this expression can be interpreted as the output of a matched filter. The core principle involves coherently accumulating the conjugate product of the observed measurements $\mathbf{Z}_k^{VFMFC}(l)$ in frequency-domain and their corresponding predicted spectra

$\mathbf{h}^H(\mathbf{X}_k)$. When a tone is present in the observed measurements, they can be coherently matched, resulting in a significant accumulation. Conversely, in the absence of a tone, the noise phase is random, preventing coherent matching and leading to a smaller accumulated value. Consequently, the magnitude of the likelihood ratio function directly reflects the likelihood of tonal presence. This concept of coherent matching finds applications in diverse domains, such as matched-field localization.

B. Order determination for the TBD model

The established TBD model incorporates $M + 1$ tonal state variables. The variable M refers to the order of the polynomial in Eq. (4) used for approximating the received signal in the time domain. An excessively large M leads to increased computational complexity. Conversely, a small M may result in significant approximation error, potentially compromising the detection performance. Therefore, determining an optimal value for the polynomial approximation order M is a critical aspect.

In the likelihood ratio function of Eq. (19), $|\mathbf{h}^H(\mathbf{X}_k) \mathbf{Z}_k^{VFMFC}(l)|$ represents the result of the coherent matching between the predicted model and the actual measurements. We define

$$G_{VFMFC} = \frac{1}{W} |\mathbf{h}^H(\mathbf{X}_k) \mathbf{Z}_k^{VFMFC}(l)|. \quad (20)$$

The theoretical coherent processing gain of G_{VFMFC} is $10 \log_{10}(WT_0)$. However, the calculation of $\mathbf{h}^H(\mathbf{X}_k)$ incorporates two steps of approximation: (1) The theoretical signal model in Eq. (2) is approximated by a M -order polynomial phase signal in Eq. (4). (2) In Eq. (6), to obtain an analytical expression for the segmented DFT spectrum of the approximated signal, terms of second order and higher in time are disregarded.

Such approximations may lead to a loss of coherent processing gain. The simulation results quantifying the gain loss incurred by the approximations in steps (1) and (2), individually and combined, for various target motion scenarios, are presented in this section. By analyzing the influence of polynomial order on the gain loss, the optimal choice for the model order is finally determined. Considering the generally low velocities of underwater targets, the probability of acceleration or deceleration is relatively low. Therefore, we limit our analysis to scenarios involving constant velocity motion, exploring constant-velocity (CV) and constant-turn (CT) scenarios.

1. Exploration under CV motion

The target trajectory of the CV motion is shown in Fig. 2. The initial distance between the target trajectory starting point and the receiver is 3.0582 km, while the closest distance during the movement is 0.7048 km. The source speed is 2.5 m/s. The source frequency is set to 391 Hz. The full

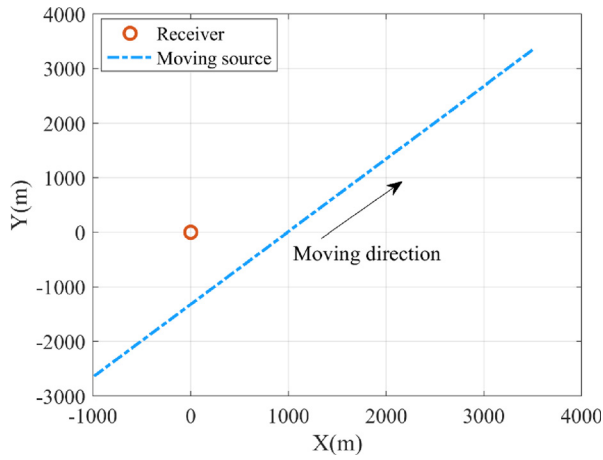


FIG. 2. The target trajectory of the CV motion.

length of simulated data is 3000 s and the frame length is 10 s.

Assume that the signal amplitude is 1, the theoretical received signal without noise is simulated according to Eq. (2). As shown in Fig. 3(a), the instantaneous frequency of the signal decreases from 391.64 to 390.36 Hz over a duration of 3000 s. The time interval between 500 and 1500 s witnesses a relatively rapid frequency change. In Fig. 3(b), the frequency variation under each batch length increases and then decreases with time, with the difference that the overall frequency variation curve is pulled up as the batch length grows. This indicates that a larger batch length places a higher demand on the ability of the algorithm to mitigate the impact of Doppler frequency shifts.

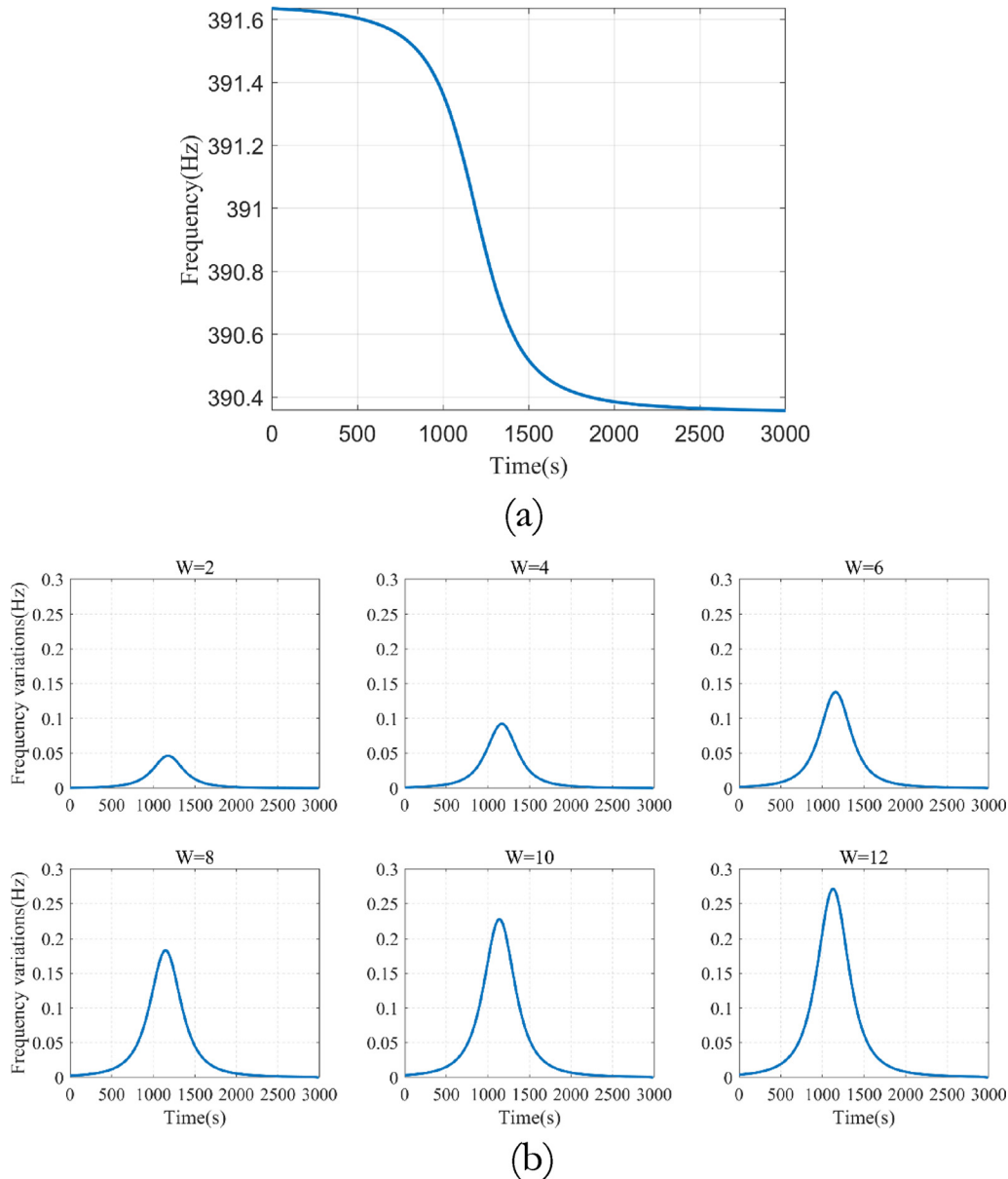
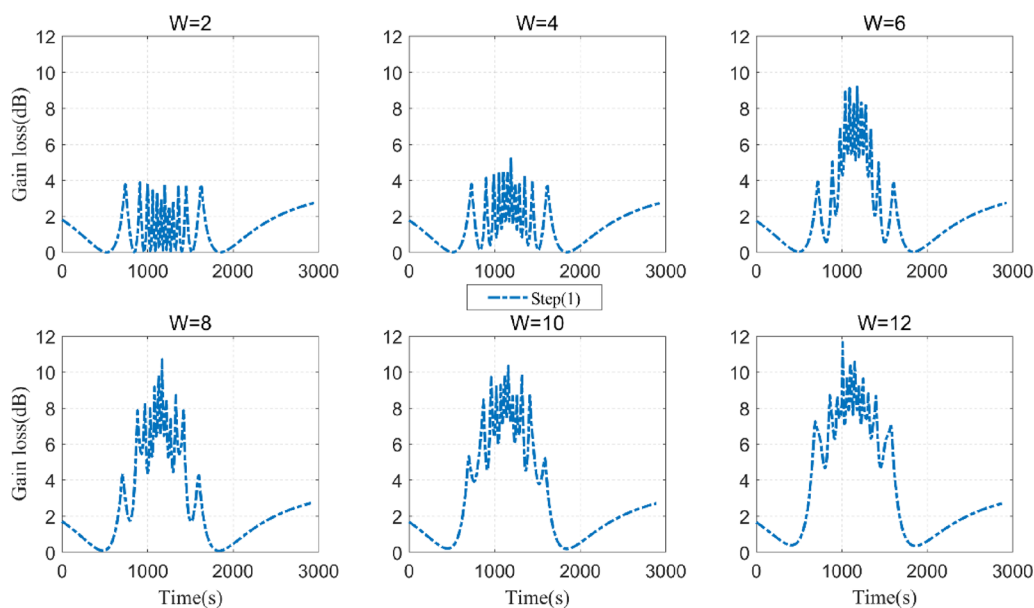
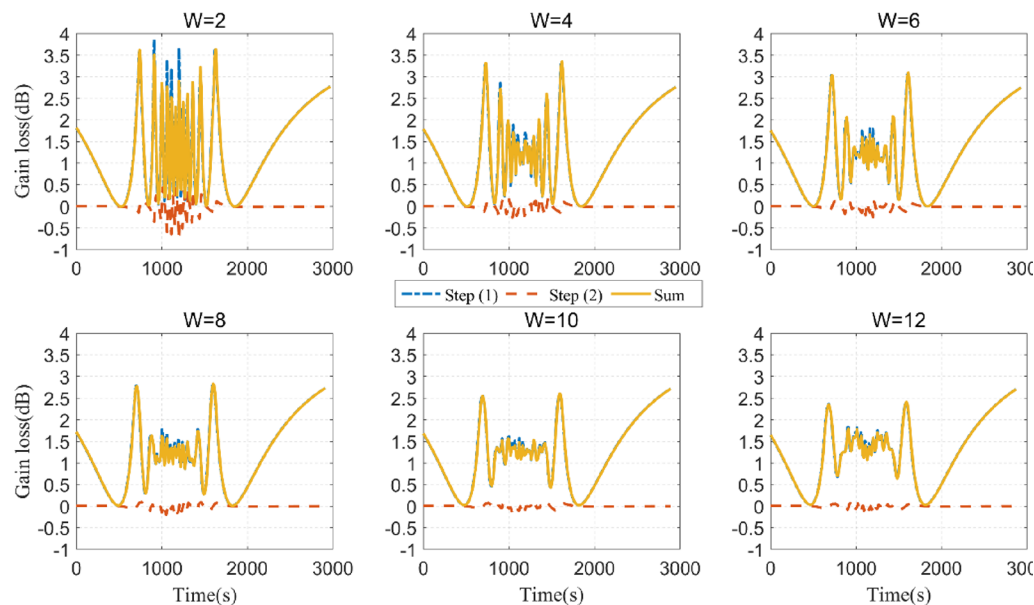


FIG. 3. Frequency variations under CV motion. (a) Curve of instantaneous frequency versus time, (b) frequency variations within different batch lengths.



(a)



(b)

FIG. 4. Gain loss for polynomial order (a) $M = 1$, (b) $M = 2$, (c) $M = 3$ under CV motion.

The coherent processing gain loss after the step (1) approximation, the step (2) approximation, and the two-step approximation for $M = 1, 2, 3$ are shown in Figs. 4(a)–4(c), respectively. The six subfigures represent the gain loss for batch lengths of $W = 2, 4, 6, 8, 10, 12$, respectively.

In Fig. 4(a), the received signal is approximated as a polynomial phase signal with $M = 1$, implying that the signal frequency is approximately constant. In this case, the approximation step (2) is absent, and the figure only presents the gain loss incurred by the approximation step (1). For the time periods with small frequency variations, i.e., 0–500 s and 1800–3000 s, the gain loss under different batch lengths remains relatively small, generally not exceeding 3 dB.

When the frequency variation is large, the gain loss remains low only when the batch length is less than four frames. This suggests that it is unwise to approximate the signal frequency as constant when it varies rapidly.

In Fig. 4(b), the polynomial order is $M = 2$. In other words, the signal frequency is approximated to vary linearly with time. It is observed that the gain loss under different batch lengths remains below 3 dB, with a slight decreasing trend as the batch length increases. The fluctuations in gain loss within time periods of small frequency variations are attributed to the picket-fence effect. When the frequency variations are significant, the gain loss is primarily dominated by the processing performance of the algorithm and

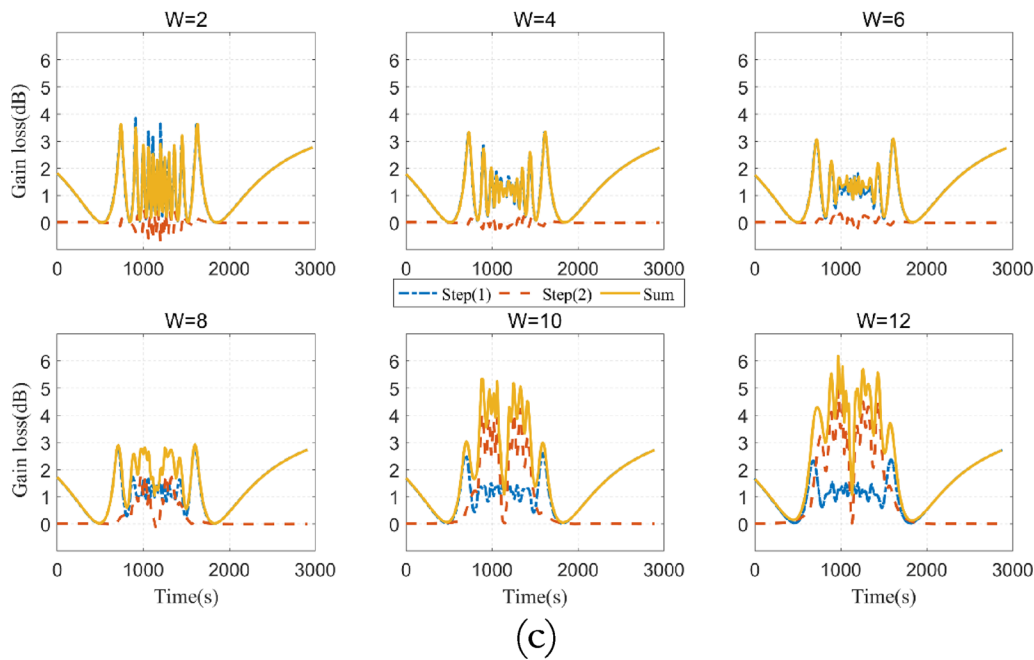


FIG. 4. (Continued)

remains relatively stable at around 1.5 dB. Furthermore, the gain loss is primarily determined by the approximation step (1), with negligible contribution from the approximation step (2). This indicates that when performing DFT on a second-order polynomial phase signal, the second-order and higher-order terms with respect to time can be disregarded, leading to a relatively straightforward analytical expression for its spectrum.

In Fig. 4(c), $M = 3$ indicates that the signal frequency is approximated as a second-order polynomial with respect to time. Comparing it with Fig. 4(b), it can be observed that the gain loss caused by the approximation step (1) for $M = 3$ under different batch lengths is slightly smaller than that for $M = 2$. For batch lengths less than eight frames, the gain loss due to the approximation step (2) for $M = 3$ is negligible, and the total gain loss remains essentially identical to that for $M = 2$. However, when the batch length exceeds eight frames, the gain loss due to the approximation step (2) for $M = 3$ increases gradually with the batch length, resulting in a significantly higher total gain loss compared to $M = 2$. This is because when the batch length is large, the quadratic term coefficient in the second-order polynomial describing the signal frequency becomes significant, and its neglect during DFT transformation leads to a substantial error in the frequency spectrum calculation.

The results indicate that for CV motion, the coherent gain loss can be controlled below 3 dB. This can be achieved by setting the model order to $M = 2$ and the batch length to 6–12 frames.

2. Exploration under CT motion

Figure 5 shows the target trajectory of the CT motion. The target initial position is at coordinates (1.5, 0.8 km), and it moves counterclockwise at a rate of $0.3^\circ/\text{s}$ for one

complete revolution. The source frequency is set to 391 Hz, the total data length is 1200 s, and the frame length is 10 s. The simulation approach for the received signal remains unchanged. The results of the frequency variations and gain loss for different model orders are shown in Figs. 6 and 7.

Figure 6(a) demonstrates that under CT motion, the signal frequency does not exhibit a monotonic decrease or increase over time. In our simulation, the frequency initially decreases, then increases, and finally decreases again, resulting in an overall “S”-shaped trend. Such a complex frequency variation will inevitably pose challenges to achieving coherent processing gain. The frequency variation at each batch length in Fig. 6(b) is relatively large at the beginning, middle, and end moments of the target motion. Especially when $W = 10, 12$, the frequency variation can reach more than 0.6 Hz.

As shown in Fig. 7(a), when $M = 1$, the gain loss for each batch length is quite severe. Figure 7(b) shows that the gain loss can be kept below 4 dB as long as $W \leq 8$ when $M = 2$. The significant curvature of the signal frequency trajectory in the time intervals of 0–250 s and 750–1200 s leads to an unacceptable level of gain loss when $W > 8$, as the approximation of the frequency trajectory as a straight line becomes insufficient. In Fig. 7(c), when $M = 3$, the gain loss incurred by the approximation step (1) is less than 4 dB for different batch lengths. However, the substantial gain loss introduced by the approximation step (2) prevents the overall gain loss from being kept within an acceptable range when the batch length falls between 6 and 12 frames.

The previous results shows that in case of CT motion, the received signal exhibits more complex frequency variations. To mitigate the gain loss introduced by model approximations, an optimal order of $M = 2$ is recommended, along with a batch length of four to eight frames.

3. Summary and practical order selection

In summary, both CV and CT motion lead to frequency variations in the received signal, with the latter potentially exhibiting more complex variations. To achieve robust coherent processing gain under diverse frequency variations while keeping the computational complexity manageable, an optimal model order of $M = 2$ and a batch length of four to eight frames are recommended for VFMFC-TBD method.

C. Implemented with improved particle filter

To address the performance degradation of the conventional particle filter algorithm caused by the mismatch between the assumed dynamic model and actual conditions under low SNR conditions, an improved particle filter algorithm is designed. By optimizing the particle distribution to enhance particle utilization efficiency, the detection and tracking performance for complex frequency-varying tones under low SNR is improved.

1. Theory of the conventional particle filter

The particle filter is an approximate solution of the recursive Bayesian theory under nonlinear non-Gaussian assumptions. The core of recursive Bayesian theory is to calculate the posterior probability density function iteratively. Normally, it consists of two stages: prediction and update.

The prediction stage involves calculating the prior probability density function (pdf) at time k with Chapman–Kolmogorov equation, which can be written in terms of the transition probability and the posterior pdf at the previous time. This is represented as

$$p(X_k | \mathbf{Z}_{1:k-1}) = \int p(X_k | X_{k-1}) \times p(X_{k-1} | \mathbf{Z}_{1:k-1}) dX_{k-1}. \quad (21)$$

The update stage involves updating the prior pdf based on the new measurement Z_k at time k via Bayesian reasoning, expressed as

$$p(X_k | \mathbf{Z}_{1:k}) = \frac{p(\mathbf{Z}_k | X_k) p(X_k | \mathbf{Z}_{1:k-1})}{p(\mathbf{Z}_k | \mathbf{Z}_{1:k-1})}. \quad (22)$$

In a particle filter, the posterior probability density function at time k is approximated by a set of random samples $\{X_k^i, i = 1, 2, \dots, N_s\}$ with associated weights $\{\omega_k^i, i = 1, 2, \dots, N_s\}$,

$$p(X_k | \mathbf{Z}_{1:k}) \approx \sum_{i=1}^{N_s} \omega_k^i \delta(X_k - X_k^i). \quad (23)$$

Therefore, the update of the posterior probability density function can be transformed into an update of the particle weights. According to the principle of sequential importance sampling (SIS),²⁴ the update equation of the particle weights is expressed as

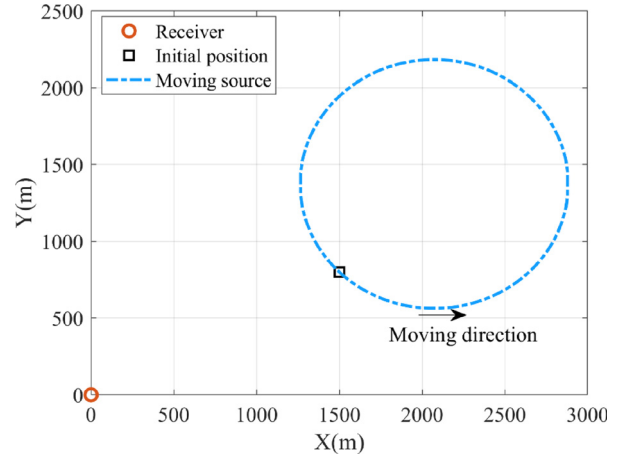


FIG. 5. The target trajectory of the CT motion.

$$\omega_k^i \propto \omega_{k-1}^i \frac{p(\mathbf{Z}_k | X_k^i) p(X_k^i | X_{k-1}^i)}{q(X_k^i | X_{k-1}^i, \mathbf{Z}_k)}, \quad (24)$$

where $q(X_k | X_{k-1}^i, \mathbf{Z}_k)$ denotes the importance density.

At time k , the state sampling for each particle is determined by the importance density, denoted as $X_k^i \sim q(X_k | X_{k-1}^i, \mathbf{Z}_k)$. The weight of each particle is calculated using Eq. (24). If the number of particles is sufficiently large, this Monte Carlo simulation method can asymptotically approach the true posterior probability density of the target. However, the number of particles is often limited in reality. As the number of iterations increases, the weights of all but a few particles can become negligible, leading to particle weight degeneracy. Resampling techniques can mitigate this problem to some extent. Therefore, a conventional particle filter framework is composed of the SIS filter and resampling methods.

2. DA-SIS

While resampling techniques can alleviate particle weight degeneracy, they may also introduce particle depletion. The loss of particle diversity inevitably affects the filtering performance, especially under low SNR conditions. If a majority of the predicted particles deviate from the true state at the current time, it can lead to target loss or erroneous tracking. To address this issue, a DA-SIS method is designed.

In Eq. (24), it is often convenient to choose the importance density to be the transition probability, i.e., $q(X_k | X_{k-1}^i, \mathbf{Z}_k) = p(X_k | X_{k-1}^i)$. So the sampling criterion of predicted particles at time k is $X_k^i \sim p(X_k | X_{k-1}^i)$. If the distribution region of the predicted particle set $\{\tilde{X}_k^i\}_{i=1}^{N_s}$ is defined as

$$\Omega(X_k) \triangleq \{X_k | p(X_k | X_{k-1}) \geq \lambda\}, \quad (25)$$

where the majority of particles can be concentrated in the high-likelihood region. Here, λ represents the pre-defined threshold for the transition probability density. The

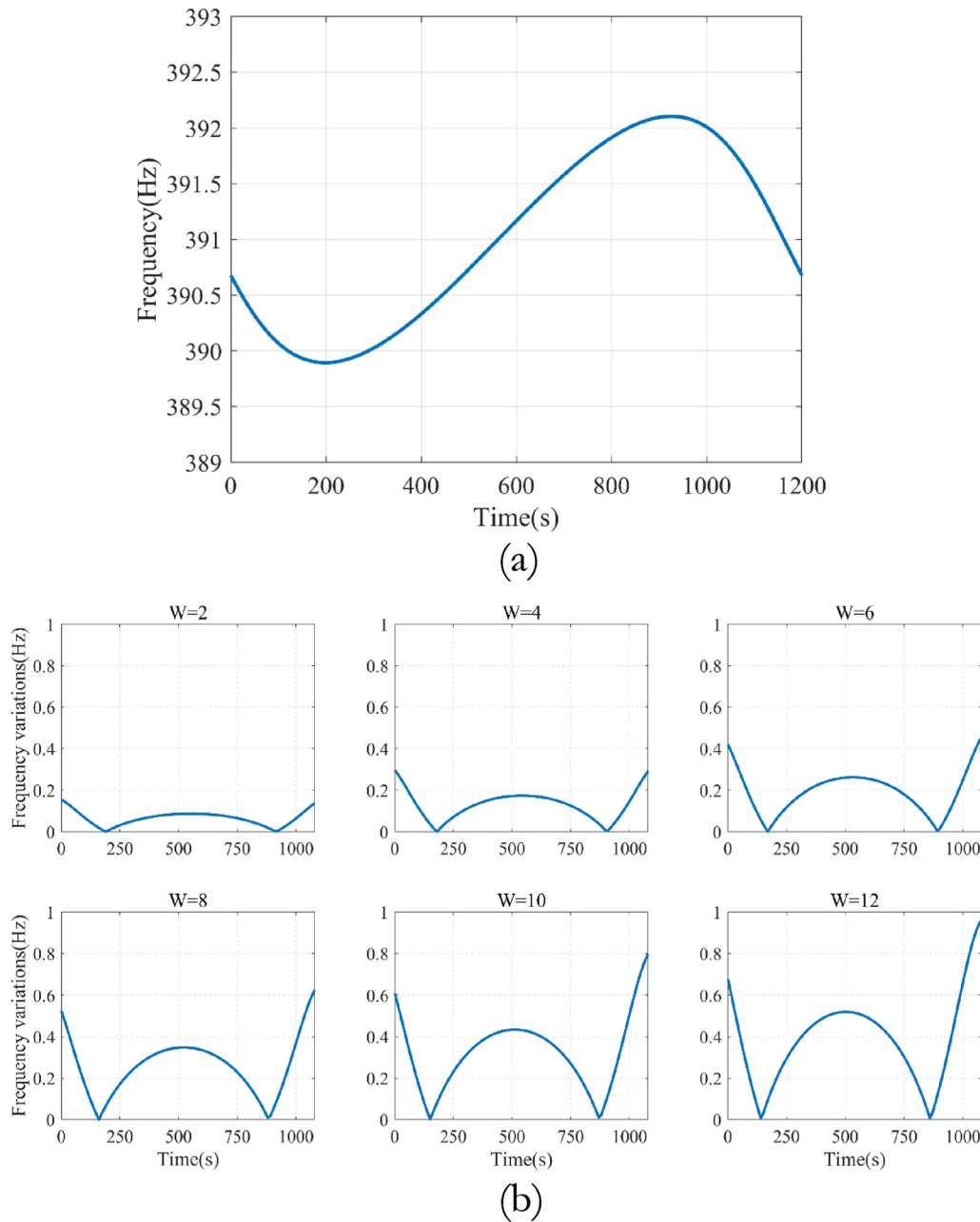


FIG. 6. Frequency variations under CT motion. (a) Curve of instantaneous frequency versus time, (b) frequency variations within different batch lengths.

advantage of this definition is that the optimized particle distribution can still represent the prior density $p(X_k|Z_{1:k-1})$ in Eq. (21) without the need for additional weight compensation steps.

According to the Markov assumption, the transition probability density derived from the dynamic model in Eq. (9) is

$$p(X_k|X_{k-1}) = \mathcal{N}(X_k; FX_{k-1}, Q_{k-1}), \quad (26)$$

where $\mathcal{N}(X; \mu, \Sigma)$ denotes the Gaussian pdf evaluated at X with mean μ and covariance Σ . The evolution of the tonal state at the current time step ideally depends solely on its state at the immediately preceding time step.

However, if the state estimates are subject to noise or information loss, relying solely on the immediately preceding estimate may be insufficient to fully characterize the true state of the system. The state estimate X_{k-1} from the previous time step may significantly deviate from the true state due to the low SNR. Furthermore, the accuracy of the transition matrix F is compromised when the tonal frequency variations do not align with the pre-defined dynamic evolution model. Both of these deviations cause the transition probability density, calculated according to Eq. (26), to diverge from the tonal true state. Consequently, the predicted particles for the current time step will be distributed in regions far from the tonal true state, thereby affecting the effectiveness of the filtering process.

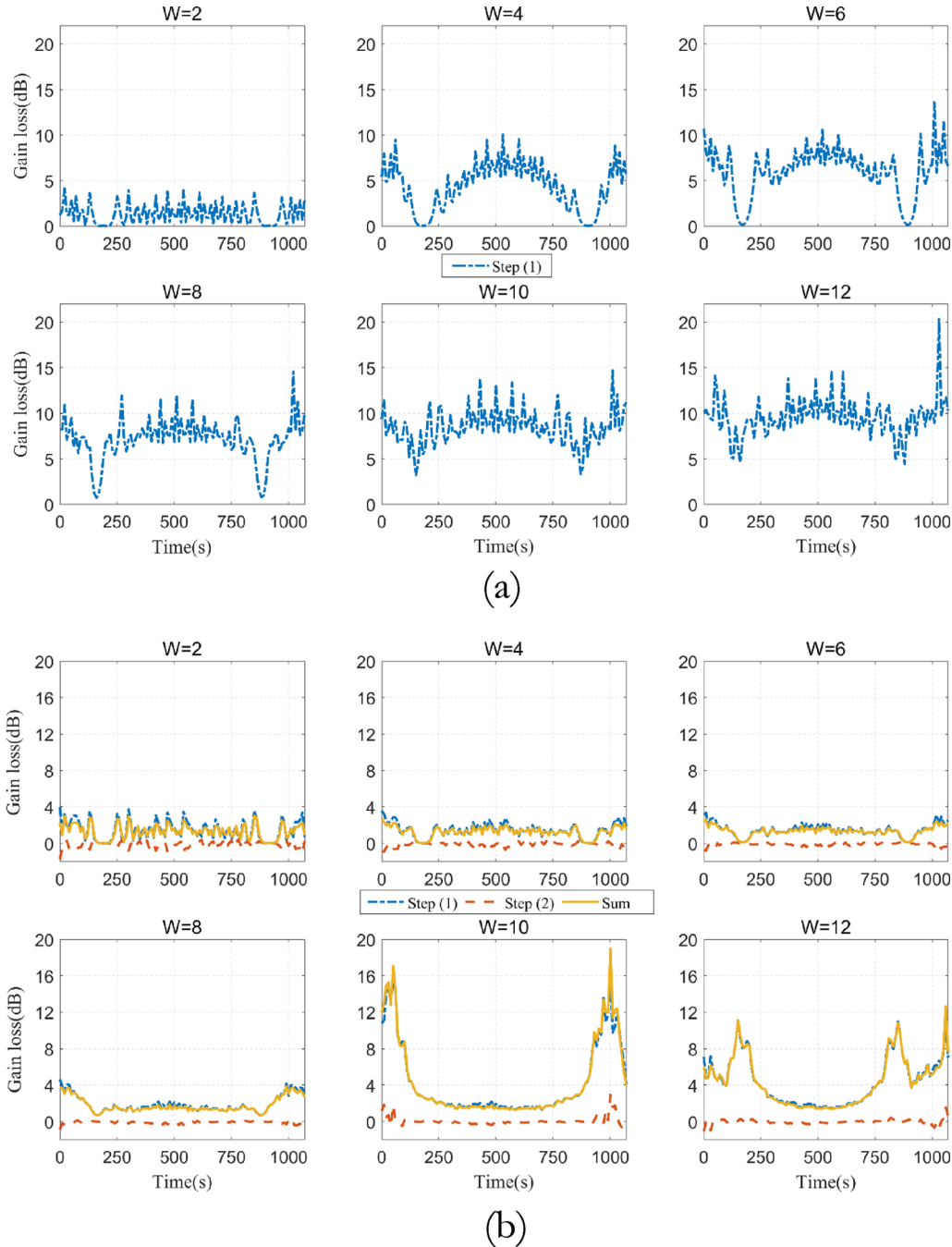


FIG. 7. Gain loss for polynomial order (a) $M = 1$, (b) $M = 2$, and (c) $M = 3$ under CT motion.

To overcome these limitations, we introduce a state sequence history into the calculation of the transition probability density, enabling a more accurate definition of the sampling region for predicted particles

$$\Omega(X_k) \triangleq \{X_k | p(X_k | X_{k-\eta:k-1}) \geq \lambda\}, \quad (27)$$

where $X_{k-\eta:k-1} = [(X_{k-\eta})^T, \dots, (X_{k-1})^T]^T$ represents the state sequence history. By adaptively predicting the probability density of the current state using a historical state sequence,³⁸ the bias caused by noise in individual state estimations can be partially compensated. This strategy

improves the conventional SIS process, enhancing the prediction accuracy of the probability density of current state and consequently optimizing particle distribution. While this improvement strategy does, from a strictly theoretical perspective, violate the Markov property, incorporating multi-step historical information bridges the gap between idealized assumptions and real-world complexity, thereby balancing theoretical rigor and practical engineering value.

If the order of the TBD model established in Sec. III A is set to 2, the state vector in Eq. (8) and the state transition matrix in Eq. (10) are simplified as follows:

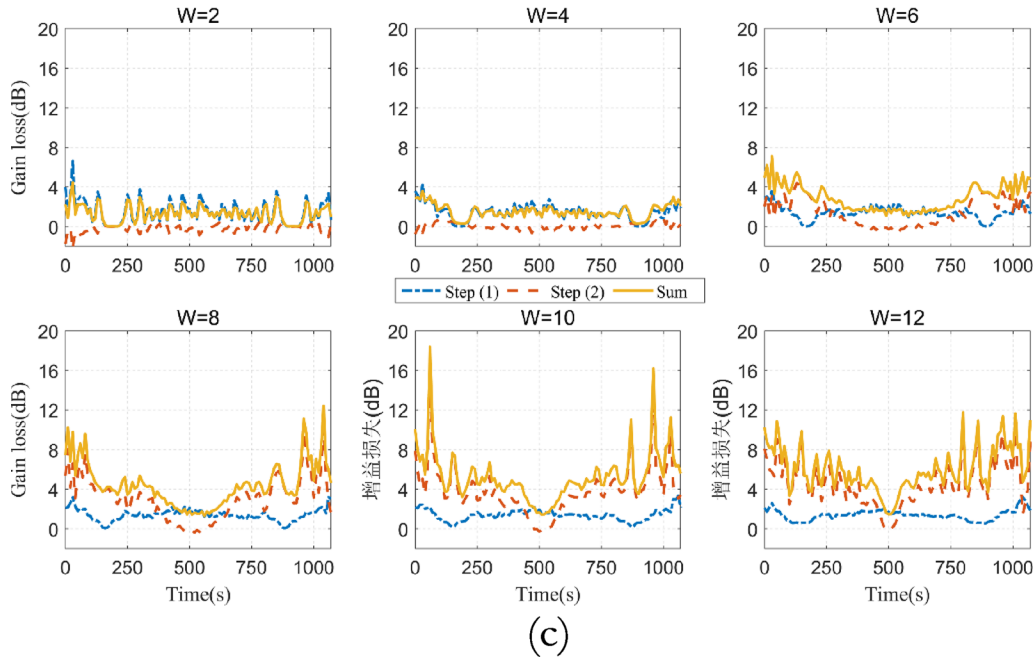


FIG. 7. (Continued)

$$X_k = [A_k, \mu_{1,k}, \mu_{2,k}]^T, \quad (28)$$

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2T_0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (29)$$

Here, $\mu_{1,k}$ and $\mu_{2,k}$ represent the coefficients resulting from approximating the received signal as a second-order polynomial phase signal. They are also related to the initial frequency and frequency rate at time k of the approximated signal.

Under the linear Gaussian assumption, the calculation process of $p(X_k|X_{k-\eta:k-1})$ is demonstrated as follows. At each time k , the relationship between the predicted target state at the current time and the state estimate from the j th past time can be expressed as the following linear equation:

$$X_j = H_{j,k}X_k + V'_{j,k}, \quad j = k - \eta, \dots, k - 1, \quad (30)$$

where the matrix $H_{j,k}$ is represented as

$$H_{j,k} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2(j-k)T_0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (31)$$

The term $V'_{j,k}$ denotes the observation error. $V'_{j,k}, j = k - \eta, \dots, k - 1$ is assumed to be mutually independent zero-mean Gaussian noise with a covariance matrix,

$$R_{j,k} = \begin{bmatrix} \beta^{j-k}\sigma_A^2 & 0 & 0 \\ 0 & \beta^{j-k}\sigma_{\mu_1}^2 & 0 \\ 0 & 0 & \beta^{j-k}\sigma_{\mu_2}^2 \end{bmatrix}, \quad (32)$$

where $0 < \beta < 1$ is a scalar coefficient, and σ_A^2 , $\sigma_{\mu_1}^2$, and $\sigma_{\mu_2}^2$ represent the variances of the observation errors for the three target states, respectively. The exponential term $j - k$ describes the increase in error as the time interval between X_k and X_j grows.

The linear Eq. (30) can be expressed in a block form,

$$X_{k-\eta:k-1} = H_{k-\eta:k-1,k}X_k + V'_{k-\eta:k-1,k}, \quad (33)$$

where

$$H_{k-\eta:k-1,k} = [(H_{k-\eta,k})^T, \dots, (H_{k-1,k})^T]^T, \quad (34)$$

$$V'_{k-\eta:k-1,k} = [(V'_{k-\eta,k})^T, \dots, (V'_{k-1,k})^T]^T, \quad (35)$$

and $V'_{k-\eta:k-1,k}$ follows a zero-mean multivariate Gaussian distribution with a covariance matrix,

$$R_{k-\eta:k-1,k} = \text{diag}(R_{k-\eta,k}, R_{k-\eta+1,k}, \dots, R_{k-1,k}), \quad (36)$$

where $\text{diag}(R_1, R_2, \dots, R_k)$ denotes a block-diagonal matrix with the elements $R_1, R_2, \dots, R_k$.

Given the state sequence history, the conditional probability density of the target state at the current time is obtained from Eq. (33) as³⁸

$$p(X_k|X_{k-\eta:k-1}) = \mathcal{N}(X_k; \tilde{X}_k, P_{k-1}), \quad (37)$$

where

$$\tilde{X}_k = P_k(H_{k-\eta:k-1,k})^T(R_{k-\eta:k-1,k})^{-1}X_{k-\eta:k-1}, \quad (38)$$

$$P_k = [(H_{k-\eta:k-1,k})^T(R_{k-\eta:k-1,k})^{-1}H_{k-\eta:k-1,k}]^{-1}. \quad (39)$$

Indeed, matrix P_k is independent of time k and can be computed off-line.

Under low SNR conditions, the tonal-existence probability may exhibit temporal instability within the historical state sequence. Specifically, at specific time indices where this probability measure falls below statistical significance thresholds, the reliability of corresponding state estimates becomes questionable. To ensure the accuracy of the conditional probability density $p(X_k|X_{k-\eta:k-1})$, it is necessary to exclude such state estimates from the calculation. The tonal state estimates within the state sequence history $X_{k-\eta:k-1}$, where the tonal-existence probability exceeds the preset threshold, δ can be represented as

$$\Pi_k = \{X_j | e_j \geq \delta, j \in N, k - \eta \leq j \leq k - 1\}. \quad (40)$$

For each target state, a polynomial curve fit is performed using the retained sequence history in set Π_k as data points. The curve fitting results, based on the least squares principle, can partially rectify state estimates with low tonal-existence probability. Consequently, substituting these fitting results into Eq. (37) for mean $\tilde{X}_k$ calculation yields a more accurate estimate of $p(X_k|X_{k-\eta:k-1})$. Therefore, the sampling region for predicted particles described by Eq. (27) can be specifically represented as

$$\begin{aligned} \Omega(A_k) &\triangleq \{X_k(1) | |X_k(1) - \tilde{X}_k(1)| \leq \lambda_1\}, \\ \Omega(\mu_{1,k}) &\triangleq \{X_k(2) | |X_k(2) - \tilde{X}_k(2)| \leq \lambda_2\}, \\ \Omega(\mu_{2,k}) &\triangleq \{X_k(3) | |X_k(3) - \tilde{X}_k(3)| \leq \lambda_3\}, \end{aligned} \quad (41)$$

where $\lambda_1 = \varepsilon_1 \sqrt{[P_k]_{1,1}}$, $\lambda_2 = \varepsilon_2 \sqrt{[P_k]_{2,2}}$, and $\lambda_3 = \varepsilon_3 \sqrt{[P_k]_{3,3}}$. The term $[P_k]_{i,j}$ represents the (i,j) th element of matrix P_k and $\varepsilon_1, \varepsilon_2, \varepsilon_3 \in (0, +\infty)$ are real-valued scaling factors.

Based on the aforementioned analysis, the sampling strategy for the predicted particle is designed as follows. For the i th newly born particle at time k , its predicted state follows a uniform distribution

$$\begin{aligned} A_k^i &\sim U(A_0 - \lambda_1, A_0 + \lambda_1), \\ \mu_{1,k}^i &\sim U(\mu_{1,0} - \lambda_2, \mu_{1,0} + \lambda_2), \\ \mu_{2,k}^i &\sim U(\mu_{2,0} - \lambda_3, \mu_{2,0} + \lambda_3), \end{aligned} \quad (42)$$

where A_0 , $\mu_{1,0}$, and $\mu_{2,0}$ denote the initial values of each tonal states. The i th continuing particle at time k are drawn from the transition probability density

$$X_k^i \sim p(X_k^i | X_{k-1}^i). \quad (43)$$

Furthermore, the particle distribution needs to be optimized based on Eq. (41). This sampling strategy ensures that the predicted particles are distributed as close as possible to the true tonal state without disrupting the prior density. However, if the sampling range of one or several state

variables has a negligible impact on algorithm performance, restricting them would increase computational time and waste resources. Therefore, the sensitivity of algorithm performance to the sampling range of each state variable will be analyzed through simulations in the following section.

3. Sensitivity analysis of state sampling range

The received signal is simulated based on Eq. (4) with $M = 2$. Considering that the signal frequency variation is solely dependent on the coefficient μ_2 , the coefficient μ_1 is set to 391 Hz. Two simulations are conducted by changing parameter μ_2 to 3×10^{-5} and 3×10^{-3} , respectively. The frame length is 10 s. For each simulation, the variation intervals for the parameters λ_1 , λ_2 , and λ_3 that determine the state sampling range are set to $\lambda_1 \in [0.001, 1]$, $\lambda_2 \in [0.01, 0.15]$ and $\lambda_3 \in [1 \times 10^{-6}, 0.1]$, respectively. Figure 8 presents the statistical average results of the tonal-existence probability obtained from 100 Monte Carlo experiments.

Comparing Figs. 8(a) and 8(b), it is evident that the tonal-existence probability is nearly insensitive to parameters λ_1 and λ_2 , remaining close to 1 under both simulation conditions. When $\mu_2 = 3 \times 10^{-5}$, the tonal frequency remains nearly constant. The tonal-existence probability decreases as μ_2 increases, suggesting that smaller values of μ_2 are more favorable for detection. When $\mu_2 = 3 \times 10^{-3}$, the tonal frequency changes relatively quickly. Only when $\lambda_3 \in [0.0035, 0.01]$ can the tonal-existence probability reach above 0.9. Therefore, the sampling range for the state variable $X_k(3)$ significantly impacts the algorithm performance and needs to be adaptively determined based on the rate of change of the tonal frequency.

In a word, the algorithm performance is most sensitive to the sampling range of state variable $X_k(3)$. To reduce the computational time, Eqs. (41) and (42) can be simplified as

$$\Omega(\mu_{2,k}) \triangleq \{X_k(3) | |X_k(3) - \tilde{X}_k(3)| \leq \lambda_3\}, \quad (44)$$

$$\mu_{2,k}^i \sim U(\mu_{2,0} - \lambda_3, \mu_{2,0} + \lambda_3). \quad (45)$$

4. Implementation steps and performance validation

By optimizing the particle distribution with online study, the improved particle filter can overcome the performance degradation caused by model mismatch. The flow chart for the proposed method is summarized in Fig. 9. For the multi-tone scenario with separate frequency components, L particle filters are simultaneously employed at L discrete frequency bins within the analysis bandwidth to provide detection and tracking results.^{25,32} However, the handling of intersecting frequency trajectories remains a limitation that requires further research.

To validate the algorithm performance, the tonal-existence probability estimated by the conventional and improved particle filter under the same simulation conditions in the previous section is calculated. The result is shown in Fig. 10.

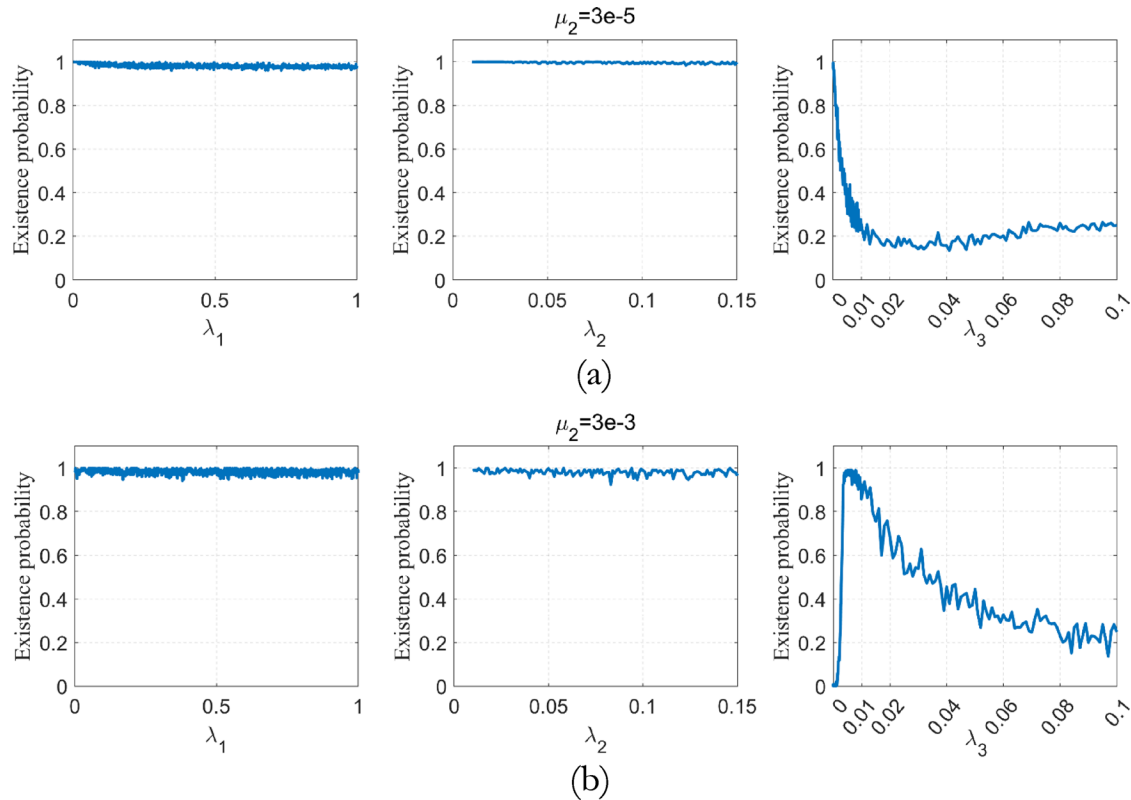


FIG. 8. The relationship between the tonal-existence probabilities and the state sampling range is depicted when (a) $\mu_2 = 3 \times 10^{-5}$ and (b) $\mu_2 = 3 \times 10^{-3}$.

The improved algorithm consistently exhibits higher existence probability across various values of λ_3 . To achieve an existence probability exceeding 0.8, the improved algorithm extends the selectable range of λ_3 by more than two-fold. When the tonal SNR is low, the performance of the particle filter becomes increasingly sensitive to particle sampling distribution. Therefore, the proposed algorithm is expected to improve the detection performance for weak tones.

To assess the probability of false alarm (P_f) for the two methods, a simulation is performed. The simulated data consists exclusively of noise. For a comprehensive evaluation, 10 000 Monte Carlo trials are conducted for each specified value of the parameter λ_3 . Target detection is declared when the estimated existence probability exceeds 0.8. The comparative results P_f for the conventional and improved particle filters are illustrated in Fig. 11.

The results presented in Fig. 11 reveal that the improved particle filter exhibits an approximate doubling of its false alarm probability. This suggests that the improved detection capability might be accompanied by an increase in false alarms. Despite this, the probability of false alarm remains bounded by 4×10^{-3} , a value considered acceptable for practical applications.

D. Analysis of computational cost

Since the covariance matrix P_k can be calculated offline, the real-time implementation capability of the VFMFC-TBD algorithm is determined by the five iterative

steps of the particle filter depicted in Fig. 9. The frame length is set to 10 s, and the batch length is six frames. The run times of the SF-TBD, MFNC-TBD, MFC-TBD, and VFMFC-TBD algorithms are investigated using a computer with an Intel Core i5-1240P CPU (1.70 GHz; Intel, Santa Clara, CA) and 16 GB RAM, running Windows 11 Home, as detailed in Table I.

The run times of Steps 1 and 3 are relatively longer. The proposed algorithm achieves a total runtime of approximately 0.0363 s, which represents an order of magnitude performance decrease compared to other TBD methods.^{25,32} However, this run time remains significantly lower than the frame length of 10 s. Furthermore, the computational cost shows no significant increase with larger batch length, making it suitable for real-time implementation.

IV. SIMULATION EXAMPLE

In this section, the performance of SF-TBD, MFNC-TBD, MFC-TBD, and VFMFC-TBD algorithms are compared through two simulation examples.

A. Simulation 1: Results of CV motion

In Simulation 1, the target moves linearly across the receiver at a constant velocity. Figure 12(a) shows the SNR of the received signal when the source level is 135 dB. The source frequency is set to 400 Hz. The frame length and batch length are 10 s and eight frames, respectively. The instantaneous frequency and the frequency variation within a single batch (each batch is time-stamped by its start time)

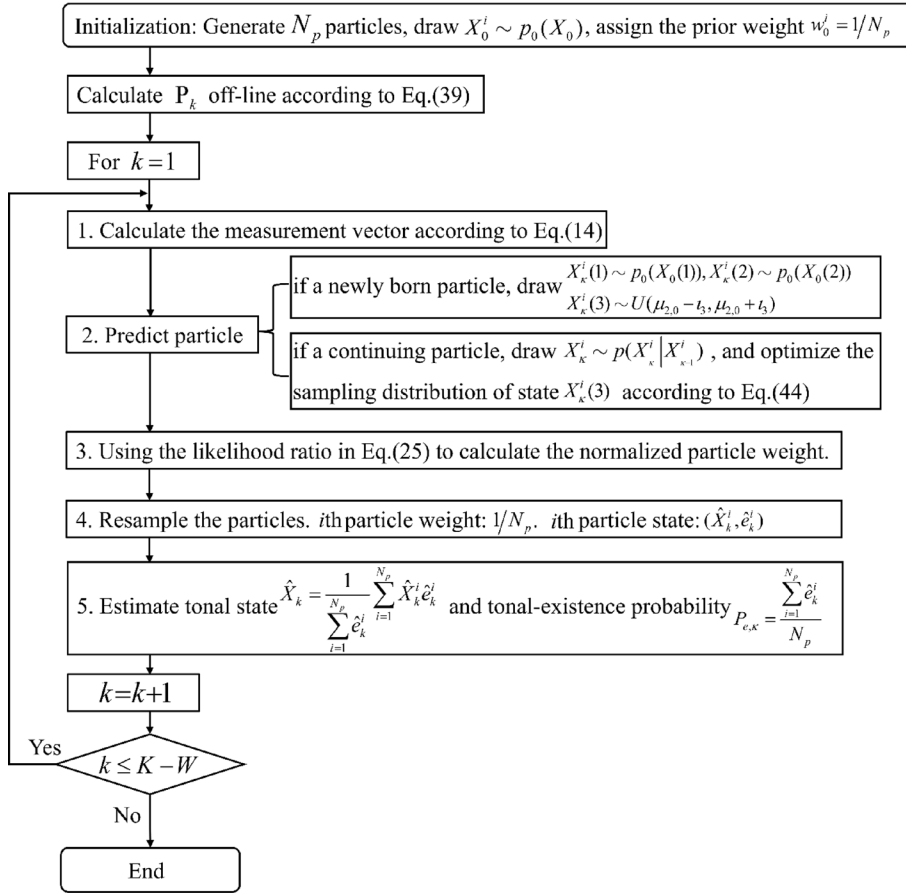


FIG. 9. Flowchart of the VFMFC-TBD method.

of the received signal are presented in Figs. 12(b) and 12(c), respectively. Within a batch, the frequency variation is the absolute difference between the instantaneous frequencies at its start and end. Since the target passes over the receiver during the observation period, approaching and then receding, the received SNR exhibits an initial increase followed by a decrease. The instantaneous frequency starts from a value greater than the source frequency, decreases to equality with it, and continues decreasing. The signal frequency variation within the single batch also follows a similar trend,

initially increasing and then decreasing, with all values exceeding the frequency resolution of 0.1 Hz.

Four TBD algorithms were employed to process the simulated signal. Based on Fig. 12(c), detection and tracking results for different frequency variations are obtained when the source level is 135 dB. Repeating this process with varying source levels and averaging the results of 100 Monte Carlo trials for each source level, we obtain the estimation results of tonal-existence probability as shown in Fig. 13, along with the frequency estimation error depicted

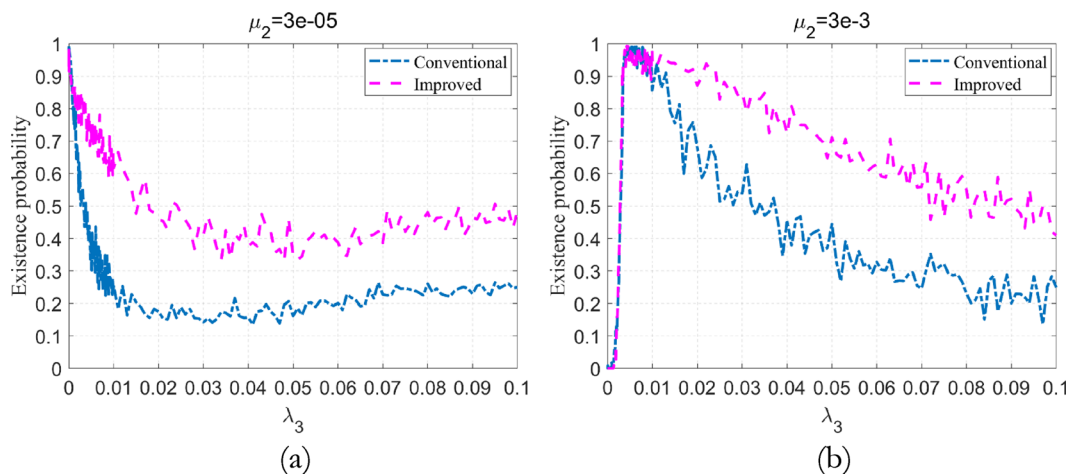


FIG. 10. Performance comparison between conventional and improved particle filter algorithm is depicted when (a) $\mu_2 = 3 \times 10^{-5}$ and (b) $\mu_2 = 3 \times 10^{-3}$.

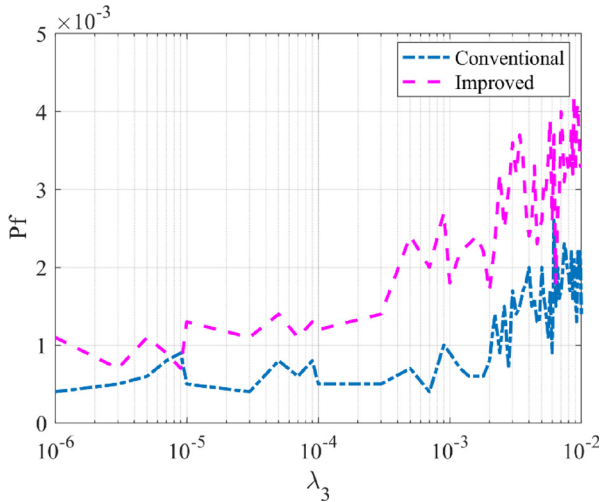


FIG. 11. The probability of false alarm of the conventional and improved particle filter methods.

in Fig. 14. It is worth noting that the results do not exhibit symmetry with respect to the frequency-shift coordinates. This can be attributed to the fact that the particle filter requires an initial period of several time frames to achieve convergence. A key observation is that lower source levels lead to a slower convergence rate.

In Fig. 13, the color bar represents the tonal-existence probability. Each subplot displays results with a similar sawtooth-like color pattern, which is caused by the picket-fence effect of the DFT transform. Comparing the results of the four TBD algorithms, the area with a tonal-existence probability greater than 0.8 decreases in the following order: VFMFC-TBD > MFC-TBD > MFNC-TBD > SF-TBD. The white lines in Fig. 13 represent contours of constant SNR, labeled -8 , -6 , -4 , -2 , 0 , and 2 dB. At the same SNR, the tonal-existence probability for both the SF-TBD and VFMFC-TBD methods appears to be largely independent of the frequency variations. However, when the signal frequency varies rapidly, the MFNC-TBD and MFC-TBD methods exhibit a lower tonal-existence probability. When the frequency shift is 0.29 Hz, the VFMFC-TBD method provides gains of approximately 4 , 3 , and 2 dB compared to the SF-TBD, MFNC-TBD, and MFC-TBD methods, respectively.

TABLE I. Evaluation of the run times for the four TBD methods (SF, MFNC, MFC, VFMFC).

	SF-TBD(s)	MFNC-TBD(s)	MFC-TBD(s)	VFMFC-TBD(s)
Step 1	1.6×10^{-3}	1.7×10^{-3}	3×10^{-3}	9.5×10^{-3}
Step 2	5.8×10^{-5}	5.8×10^{-5}	5.9×10^{-5}	9.8×10^{-4}
Step 3	7.2×10^{-4}	8.7×10^{-4}	1.6×10^{-3}	2.57×10^{-2}
Step 4	7.5×10^{-5}	7.4×10^{-5}	7.3×10^{-5}	7.4×10^{-5}
Step 5	1.7×10^{-6}	1.6×10^{-6}	1.7×10^{-6}	1.6×10^{-6}
Total	0.0025	0.0027	0.0048	0.0363

It can be observed that the MFNC-TBD method consistently exhibits larger frequency estimation errors across different SNRs, rarely achieving errors below 0.02 Hz. The MFC-TBD method shows slightly better performance with fewer instances of errors exceeding 0.08 Hz. The SF-TBD method outperforms the previous two, maintaining frequency errors mostly below 0.04 Hz. The VFMFC-TBD method demonstrates the most optimal performance. While the estimated error slightly increases with decreasing SNR, it remains below 0.02 Hz for most cases, and the error remains largely unaffected by the frequency variations.

To facilitate a quantitative comparison of the detection performance of various TBD methods, repeated trials are performed at multiple source levels. Detection thresholds are computed using the Monte Carlo method, based on a predetermined probability of false alarm. A target detection is considered successful if the estimated tonal-existence probability surpassed the detection threshold and the estimated frequency error remains below 0.1 Hz. Figure 15 presents the detection probability versus SNR curves for different TBD methods under a frequency shift of approximately 0.29 Hz.

The results presented in Fig. 15 demonstrate that the proposed VFMFC-TBD method achieves superior detection probabilities compared to other methods across varying probabilities of false alarm. This suggests that the method effectively balances a high tonal-existence probability with a low frequency estimation error. The SF-TBD method exhibited the second-highest performance. In contrast, the MFNC-TBD and MFC-TBD methods, hampered by their larger frequency estimation errors, fail to achieve high detection probabilities even at high SNRs.

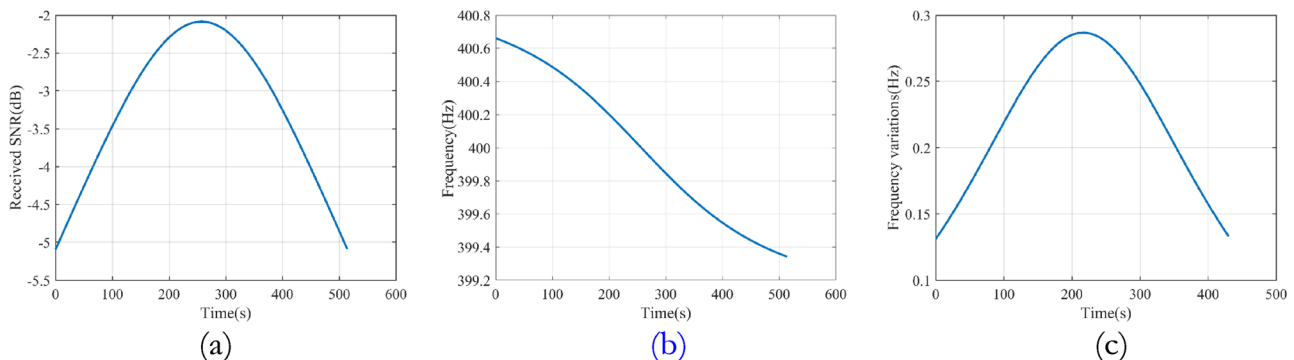


FIG. 12. Characteristics of the simulated signal in Simulation 1. (a) SNR of the received signal. (b) Frequency variation of the received signal. (c) Frequency variations within a single batch at different time instants.

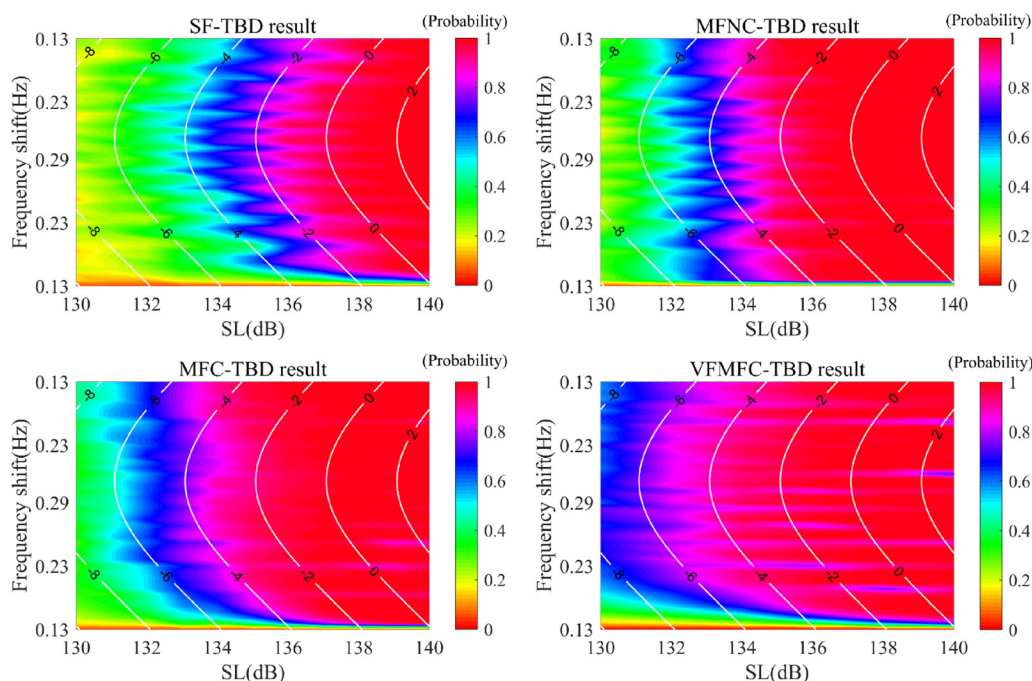


FIG. 13. Existence probability results of the SF-TBD, MFNC-TBD, MFC-TBD, and VFMFC-TBD algorithms in Simulation 1.

In conclusion, the performance of the VFMFC-TBD method is primarily influenced by the SNR and exhibits limited sensitivity to the rate of change of the signal frequency. This method can achieve high detection performance while simultaneously improving frequency estimation accuracy.

B. Simulation 2: Results of CV and CT motions

In Simulation 2, the target motion mode and parameters are listed in Table II. The target initial velocity is

$v_x = 3\text{m/s}$, $v_y = -3\text{m/s}$. The target trajectory and its distance to the receiver are shown in Fig. 16.

The received signal is simulated according to Eq. (2). Assuming a source level of 136 dB, Fig. 17 illustrates the variations in SNR and tonal instantaneous frequency. As seen in Fig. 17(a), the tonal SNR remains below 4 dB within the time intervals of 0–330 s and 1690–2290 s, making detection more challenging. The tonal trajectory in Fig. 17(b) exhibits multiple inflection points and a generally rapid rate of change, imposing a higher performance requirement on the algorithm.

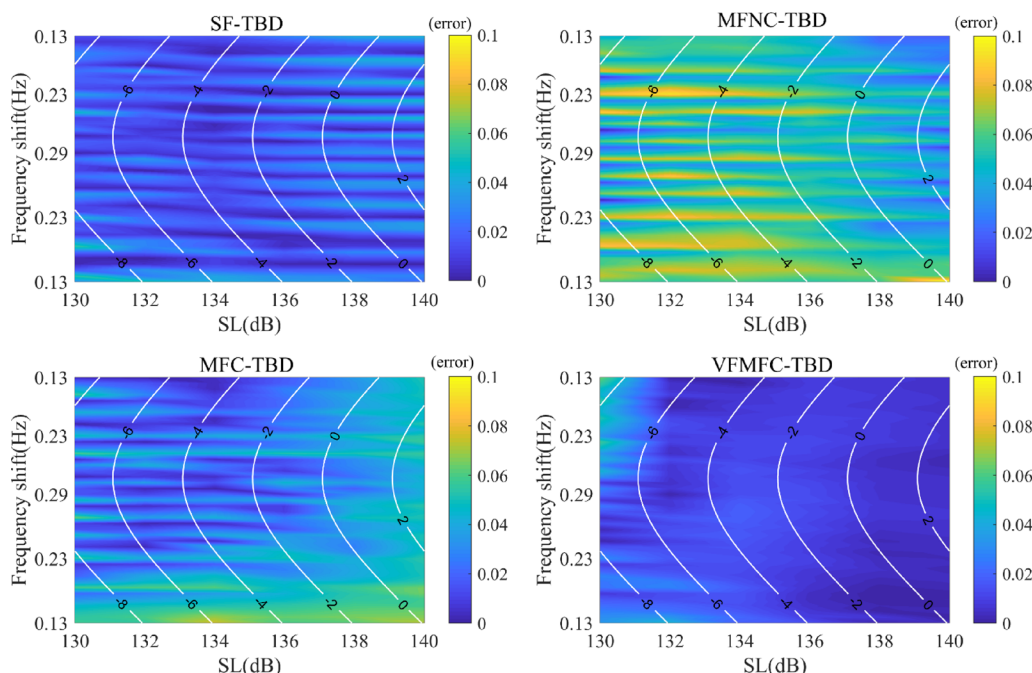


FIG. 14. Error of the frequency estimates obtained by SF-TBD, MFNC-TBD, MFC-TBD, and VFMFC-TBD algorithms in Simulation 1.

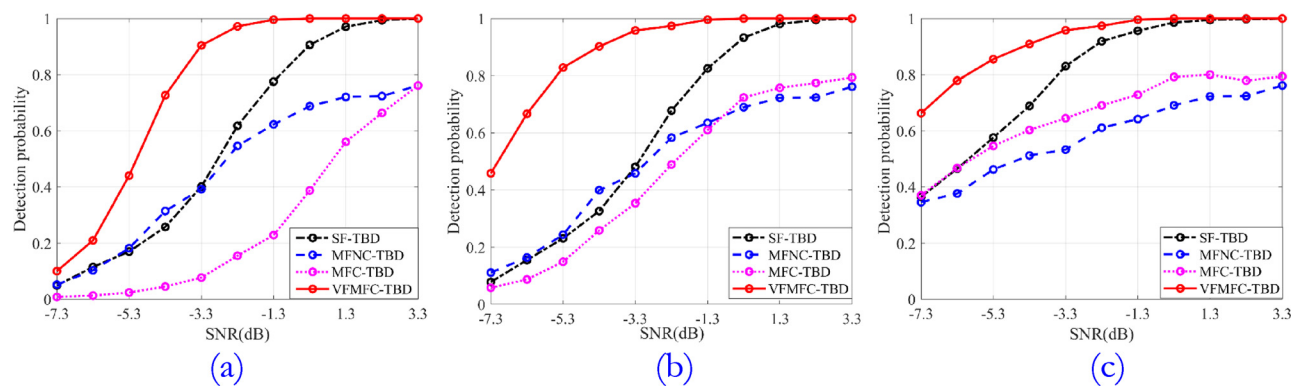


FIG. 15. Detection probabilities of the SF-TBD, MFNC-TBD, MFC-TBD, and VFMFC-TBD methods at false alarm probabilities of (a) 0.01%, (b) 0.1%, and (c) 1% when the frequency shift is about 0.29 Hz.

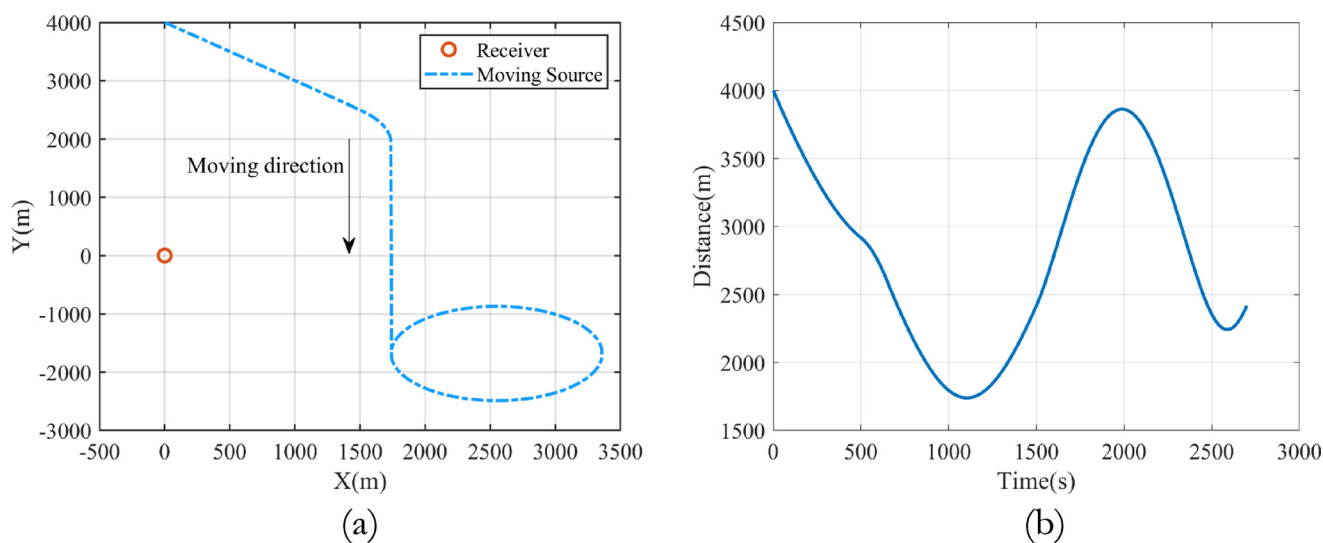


FIG. 16. Target motion scenario in Simulation 2. (a) Target trajectory, (b) distance between the target and receiver.

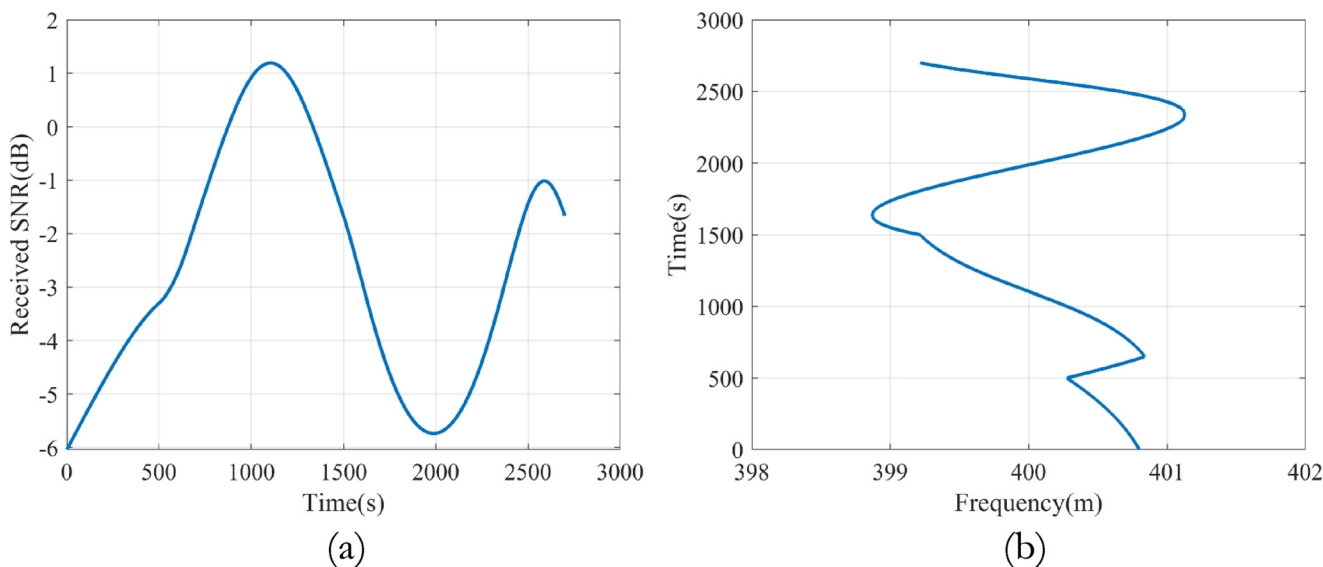


FIG. 17. Characteristics of the simulated signal in Simulation 2. (a) SNR and (b) Instantaneous frequency of the received signal.

TABLE II. Target motion mode and parameters.

	CV	CT	CV	CT
Time (s)	0–500	500–650	650–1500	1500–2700
$\omega(^{\circ}/s)$	0	−0.3	0	0.3

With a frame length of 10 s and a batch length of six frames, the processing results are shown in Fig. 18. The LOFAR in Fig. 18(a) indicates that the tonal SNR is relatively low around 2000 s, making its trajectory barely discernible. Comparing the results of the four TBD algorithms,

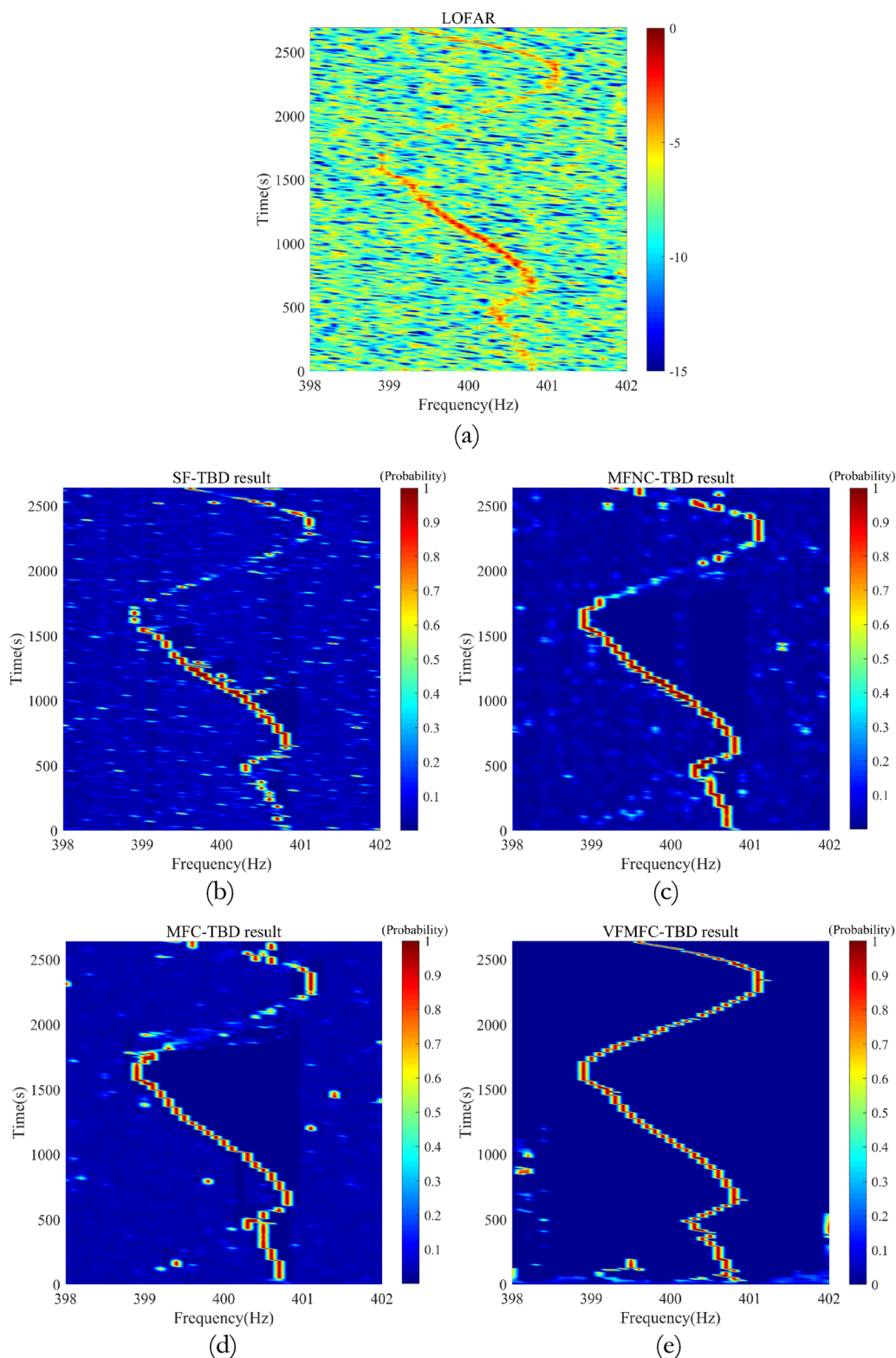


FIG. 18. Processing results of Simulation 2. (a) LOFAR of the simulated signal, (b) SF-TBD, (c) MFNC-TBD, (d) MFC-TBD, (e) VFMFC-TBD.

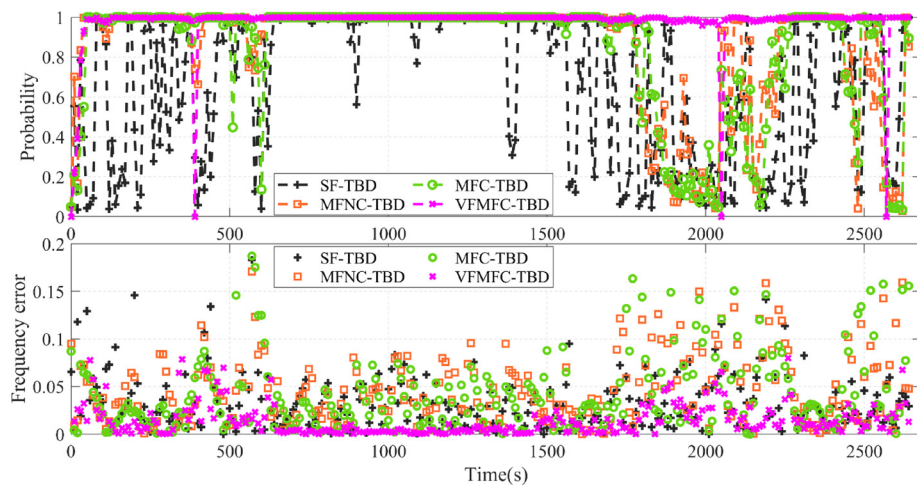


FIG. 19. Comparison of existence probability and frequency error estimated by the four TBD algorithms in Simulation 2.

it is evident that SF-TBD exhibits the poorest performance. MFNC-TBD and MFC-TBD perform similarly, failing to detect the target accurately around 2000 s and after 2500 s. VFMFC-TBD demonstrates the most robust performance, consistently detecting the tone across nearly all time intervals.

The tonal-existence probability and frequency estimation error obtained by different algorithms are compared in Fig. 19. The existence probability curve of the SF-TBD method exhibits significant fluctuations. MFNC-TBD method, being insensitive to phase information, performs slightly better than MFC-TBD method during time periods with rapid frequency changes. The VFMFC-TBD method

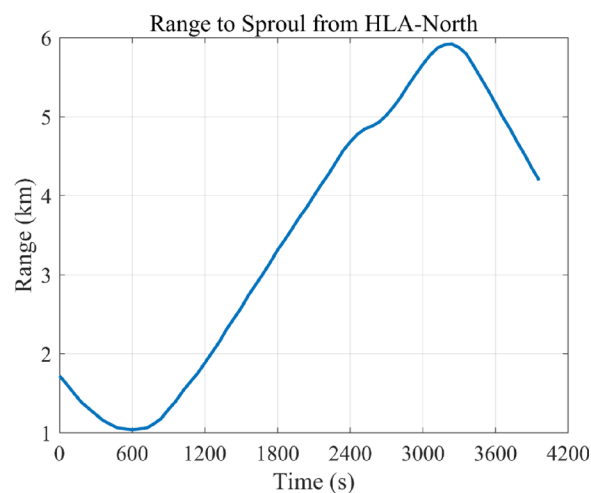


FIG. 21. Range to the target from HLA North array.

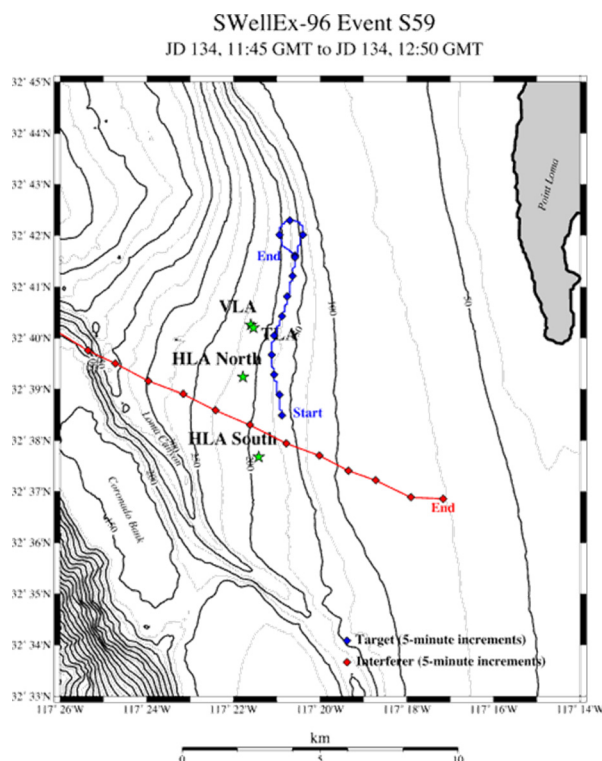


FIG. 20. Target motion trajectory (blue line) of S59 event in SWellEx-96 experiment.

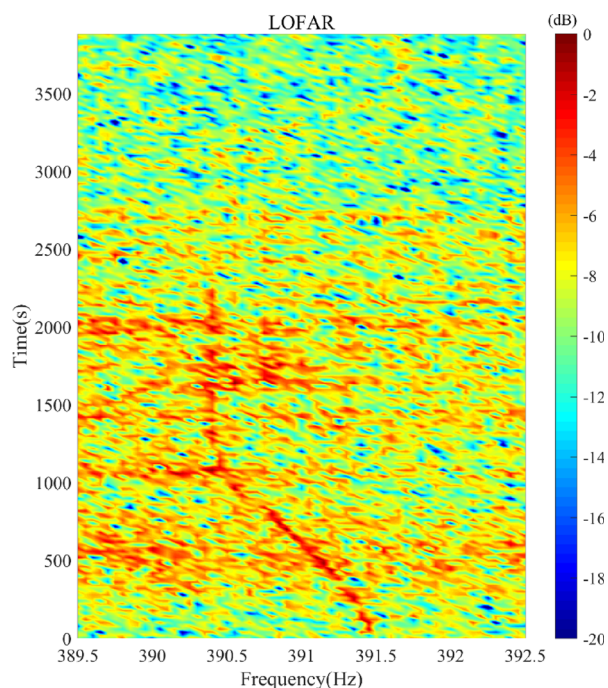


FIG. 22. LOFAR of the received data from HLA North array.

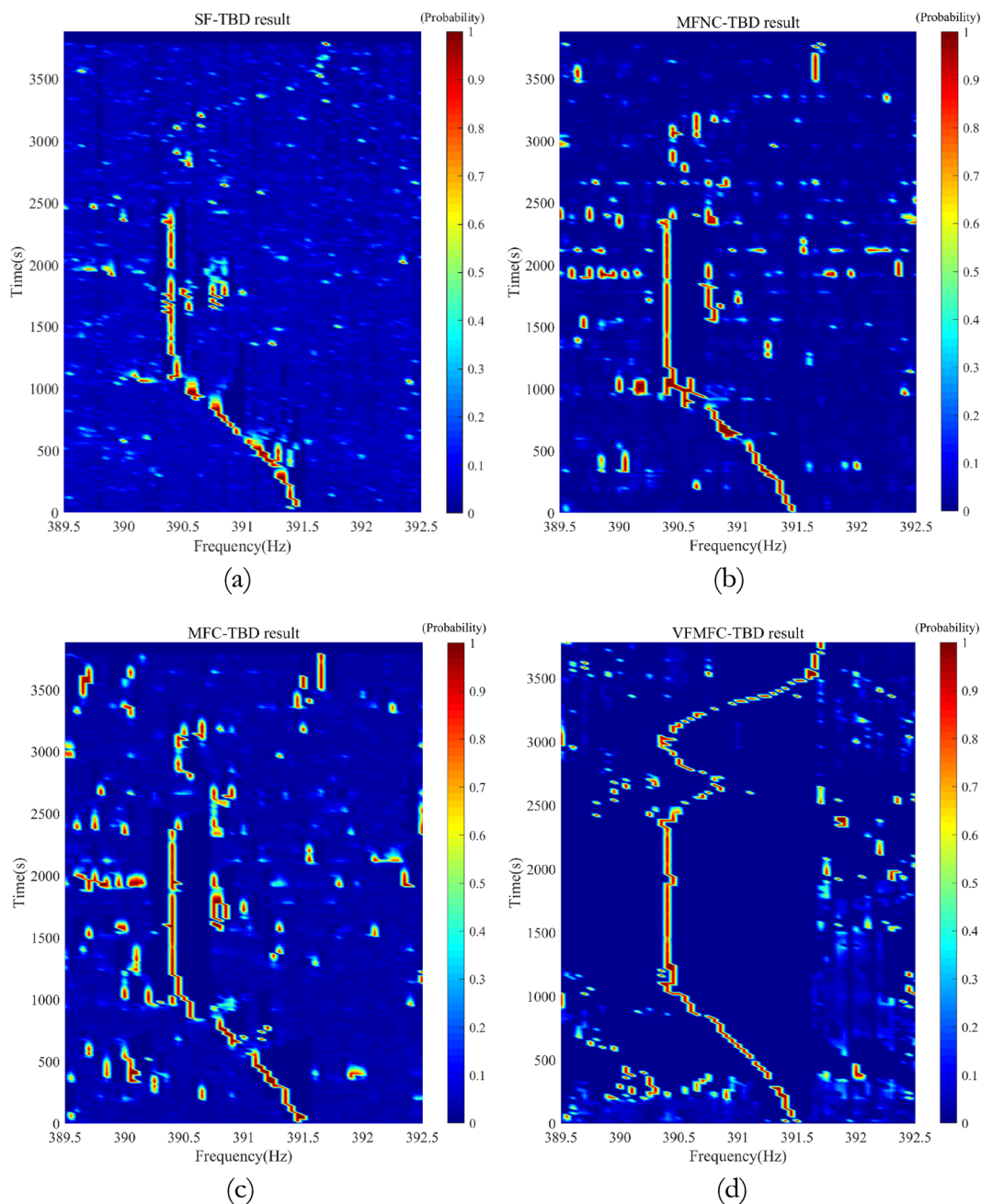


FIG. 23. Detection and tracking results of the received data. (a) SF-TBD, (b) MFNC-TBD, (c) MFC-TBD, (d) VFMFC-TBD.

demonstrates a high existence probability, except at certain points in time. Additionally, VFMFC-TBD consistently maintains a low frequency estimation error, surpassing the performance of the other three methods.

V. SEA TRIAL RESULTS

The SwellEx-96 experiment was conducted in May 1996 near San Diego, CA. In the S59 event, the target trajectory is depicted by the blue line in Fig. 20. The target vessel, carrying two acoustic sources, began its journey at the “Start” point and traveled northward at a constant speed of 2.5 m/s. In the final 20 min, the target vessel turned towards

the harbor, completing a circular movement before concluding its journey at the “End” point. The sources emitted multiple tonal signals with frequencies ranging from 49 to 400 Hz. The effectiveness of the algorithm is validated using the data recorded by the horizontal line array (HLA North). Figure 21 illustrates the distance between the target source and the HLA North array. During the time interval from 40 to 65 min, their distance is relatively large, leading to a significant decrease in the SNR of the recorded data.

The recorded data from one element of the HLA North array is processed. The frame length is set to 20 s. The LOFAR result for a tone with a source frequency of 391 Hz and a source level of 132 dB is presented in Fig. 22. It can

be observed that within the time interval of 0–2300 s, the tonal frequency initially decreases and then remains constant, resulting in a relatively clear trajectory. After 2300 s, the tonal SNR decreases significantly, making it difficult to observe the trajectory. With a batch length of six frames, the TBD results of the four algorithms are shown in Fig. 23.

In Fig. 23, all four methods successfully detect the tone within the 0–2300 s. However, for the 2300–3900 s, where the SNR is lower and the tonal frequency changes more rapidly, SF-TBD only detects the tone sporadically between 2800 and 3900 s, completely missing the trajectory between 2300 and 2800 s. MFNC-TBD and MFC-TBD perform similarly, detecting the tone only around 3000 s and 3500 s with slight deviations in the displayed trajectory compared to the actual trajectory. VFMFC-TBD successfully detects the tone throughout the duration, presenting a clear and distinct tonal trajectory.

VI. CONCLUSION

To enhance the detection performance of tones with complex frequency variations under low SNR conditions, a variable frequency-based multi-frame coherent track-before-detect method (VFMFC-TBD) is proposed and evaluated in this study. Our analysis indicates that a key limitation of existing TBD methods lies in the assumption of tonal frequency stability, which is often unrealistic in many target motion scenarios. The newly established measurement model allows for frequency shifts of the tone across different frequency bins. The matched filtering term in the derived likelihood ratio function incorporates inter-frame phase information, achieving a coherent integration gain. Furthermore, the designed DA-SIS method can adapt to tonal frequency variations, reducing the risk of missed detection due to model mismatch. Simulation results demonstrate that the proposed VFMFC-TBD method provides approximately 2–4 dB gain compared to other TBD methods. Additionally, it can significantly reduce frequency estimation errors. Experimental data processing results also demonstrate that the proposed method effectively detects tonal signals submerged in noise, exhibiting significant superiority.

In this study, the Fourier transform is utilized to linearly approximate the received nonlinear signals, facilitating the establishment of the TBD model. However, this linear-based processing can lead to specific gain losses and potentially increase the risk of missing targets in complex real-world ocean environments. We are interested in exploring nonlinear transform methods in the future to further enhance the detection performance of weak tones with time-varying frequencies.

The proposed method focuses on solving the problem of tone detection and tracking where the frequency changes deterministically. However, in complex marine environments, factors such as source frequency instability, randomly fluctuating sea surfaces, turbulence, and random disturbances in the propagation channel can cause random

frequency fluctuations. Future research can explore frequency modeling methods for these complex scenarios, thereby further expanding the applicability of this method.

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AUTHOR DECLARATIONS

Conflict of interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available at <https://swellex96.ucsd.edu/>.

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